

# Quantum Readout from Observer-Relative Lossy Descent

Tau Core Foundation Paper VI: A finite conditional quantum terminal

Jozsef Olcsak

Last revised: 3 September 2026

## Abstract

We ask which additional structures are required for an observer-relative, information-losing descent from an atemporal morphological body to support a quantum terminal rather than ordinary classical coarse graining. First, a finite countermodel proves that resolution loss, non-injectivity and stable record classes alone imply neither complex amplitudes nor interference, Born probabilities, entanglement or no-signalling. We then state a finite conditional source packet. A positive quotient metric and a compatible nondegenerate two-form construct a complex Hilbert carrier when the associated endomorphism has one common complex scale. A commuting self-adjoint generator gives unitary transport; positive normalized effects give the trace Born rule; and an explicitly selected tensor product gives local-CPTP no-signalling.

We further derive a local observer-resolution window: a nonzero continuous precursor becomes a distinct record only when its complete record jet exits a source-owned stable quantizer cell before the local geometric approximation fails. On a constant-rank access stratum, an exhaustive complete effect–Choi instrument detects every nonzero outward Kato projector jet, although instantaneous probabilities and incomplete record families may remain blind. Finally, simultaneous Parent record maps give nested observer-relative history fibres. A common pre-choice preparation and a normalized later instrument preserve the earlier marginal, so delayed-choice conditioning refines histories without rewriting a Parent past or enabling retro-signalling.

Inside an adopted enriched Tau completion, these structures are represented on one normalized complex matrix block with a common action unit. We prove finite record-separation and tensor-factorization criteria, distinguish exact logical recovery from dephasing, and give an approximate-recovery bound. The standard free algebraic-QFT and perturbative causal-QFT constructions are used only as conditional fixed-background limits. None of these standard quantum results is presented as a Tau-specific deviation.

The proposed discriminator instead compares two independently reconstructed quantities on the same occupied record pair: a normalized body-action separation and its Uhlmann chord. The completion requires them to agree, with a certified recovery-error envelope. A public-data audit finds no carrier on which the action-side quantity is independently observable, so the discriminator is not yet executable. We also prove that the present narrow base–seed reduct does not select the full joint source packet. Consequently the paper establishes a mathematically explicit conditional quantum terminal, not a quantum theory derived from the unrestricted physical parent law; it does not derive the SI value of  $\hbar$ , physical outcome selection, nonperturbative QFT, quantized geometry or the Standard Model. In particular, the quantum projective measure is not identified with body four-volume by rank equality alone; a full source-owned projective/body connector remains an independent premise. A braid-holonomy audit similarly gives an exact conditional anyon-readout criterion but proves that the current source does not select its topology or representation. The same audit isolates a source-frozen three-mode susceptibility target; it is a conditional precursor, not a reported experimental prediction. Finally, the paper supplies the quantum operational obligations for observer-dependent cross-terminal mixing: information-level role exchange does not identify a contextual quantum packet with a classical noncontextual gravity terminal.

**Inherited morphology terminology.** The input body is the complete *stabilized morphological response configuration*  $M_\tau^*$  selected upstream from a base carrier and its base-readable seed/load pattern; “morphological body” is shorthand. This paper neither identifies the body with the bare base nor assumes a literal formation trajectory. Its occurrence and quantum instruments are post-body readouts and do not retrospectively supply parent time. Under MRC-DEF2, parent-side objects are typed pregeometric relational candidates, not particles or apparatuses already embedded in recovered spacetime; those ordinary objects occur only in downstream body/readout roles.

**Dependency and ownership.** Inputs are: the atemporal body and observer-access architecture (Paper I), typed finite source support (Paper II), recovery-correctable record transport (Paper III), the common-source occupation boundary (Paper IV), and stable observer quantization plus a calibrated terminal clock (Paper V). This paper owns the loss-only no-go, the finite metric-symplectic quantum completion, Born/interference and no-signalling terminals, and the quantum/classical status ledger. It also owns the local quantizer-resolution window, the outward-access/exhaustive-Choi detection theorem, the atemporal history-refinement boundary, the recovery-based factor selector, its approximate error bound and the central-chirality terminal-blindness theorem. It neither re-proves nor silently strengthens its inputs. The downstream scalar-15/type-II mass and RG specialization is owned by Paper VII-C; it does not replace the observer quantizer or the quantum source packet proved here. Paper V’s temporal-ontology underdetermination is inherited unchanged: when the quantum and record terminals factor through the observer-accessible restriction of one completed body, they cannot distinguish that body from a coherent staged realization with the same restrictions. Quantum ambiguity, unitary transport or outcome records therefore may not be used backwards to infer literal parent morphogenesis or an external parent time.

**Quantum specialization of the unified terminal atlas.** For the shared preterminal descriptor  $\Xi_{OS}$  owned upstream, the quantum terminal has the inherited form

$$Y_{OS}^{(Q)} = Q_{OS}^{(Q)} \circ T_{OS}^{(Q)} \circ \Xi_{OS} \circ A_{OS}[M_\tau]. \quad (1)$$

Its distinctive content is not a second source but the phase/effect, composition, tensor and stable-resolution structures retained by  $T_{OS}^{(Q)}$  and  $Q_{OS}^{(Q)}$ . A hybrid terminal is likewise a multi-leg descendant,

$$Y_{OS}^{(H)} = Q_{OS}^{(H)} \circ H_{OS} \left( T_{OS}^{(a_1)} \Xi_{OS}, \dots, T_{OS}^{(a_k)} \Xi_{OS} \right), \quad (2)$$

not an independent parent channel. Common ancestry does not identify contextual quantum structure with a classical noncontextual terminal: operational equivalence still requires preservation of composition, effects, phases, probabilities and no-signalling structure. A terminal varying along a  $\Xi_{OS}$ -fibre would instead be a common-ancestor bypass and would demote the declared atlas-completeness claim.

The quantum terminal also inherits, rather than creates, the visible four-dimensional carrier. A possible upstream representation uses a common body-side  $*$ -derivation family  $\mathfrak{d}_\tau$ , a source-owned access restriction  $R_{OS}^\mathfrak{d}$  and

$$\Phi_{OS,x} = \tilde{\Phi}_{OS,x} \circ R_{OS}^\mathfrak{d}.$$

Neither finite resolution, non-injectivity, phase retention nor the Born packet forces rank  $\Phi_{OS,x} = 4$ , its integrability, or a Lorentz-conformal realization. Conversely, a regular rank-four quotient does not supply coherent composition, effects, tensor structure or no-signalling. Quantum recovery and 4D reconstruction are therefore distinct conditional gates joined only by a source-owned common carrier.

**Atemporal parent-realization terminology.** Because the Tau parent is atemporal, phrases such as “Nature selects” do not denote an external agent or a later choice. Their precise meaning is the internal physical implication

$$(B_{\tau}^{\text{phys}}, s_U, \mathcal{L}_{\text{parent}}) \implies \mathfrak{P},$$

together with occupation of the support of the resulting packet  $\mathfrak{P}$ . A theorem inside a declared completion proves only class-relative realization. It does not prove this unrestricted base-seed arrow, and agreement of finitely many terminal readouts cannot by itself establish ambient-source exhaustivity. Current-facing statements therefore use *parent-law realization* and *physical occupation*; any retained legacy selection language is shorthand for this same distinction.

**Exterior source-grammar refinement.** Let  $V_s$  be an oriented primitive possibility space. If primitive directions are multiplicity-free, so that  $v^2 = 0$ , bilinear polarization gives  $uv = -vu$ . With a unit and no additional source relation, the complete finite grammar is therefore  $\Lambda^\bullet V_s$ , and the empty word is its distinguished root. A positive base metric then supplies insertion and contraction operators and hence the finite Clifford/cyclic carrier used below. Tensor and symmetric grammars are explicit controls: primitive inventory and composition alone do not select the exterior branch. The result is a conditional source-class theorem, not proof that Nature occupies that class.

## 1 Inherited proof hierarchy

Paper VI is a consolidation and terminal-recovery paper, not a restart of the quantum branch. A corpus-wide audit of the theory hub and Papers I–V gives the following inherited ladder.

| Level             | Inherited result                                                                                                                                            | Status used here                                                                                            |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Algebraic         | A full-role operator square<br>$C(v)^2 = K_*(v, v)I$ polarizes to the Clifford relations                                                                    | Exact implication                                                                                           |
| Representation    | Seven P3 generators admit the finite packet $\text{Cl}_7(\mathbb{C}) \simeq M_8(\mathbb{C}) \oplus M_8(\mathbb{C})$                                         | Constructed conditional representation                                                                      |
| Completion        | The graph-normal P3 handoff realizes the operator square with one action unit and zero body Schur remainder                                                 | Adopted minimal post-body completion                                                                        |
| Record/terminal   | Full S4 effects and the S5 square-root terminal supply the sharp spectral instruments; the matrix block supplies metric, phase, unitary and Born structures | Conditional recovery inside the enriched MVP                                                                |
| Field limit       | Globally hyperbolic descent, Hadamard preparation and local causal factorization supply free and perturbative interacting QFT structure                     | Conditional standard recovery; interaction content and nonperturbative limit open                           |
| Physical source   | The current narrow parent has no typed full-role incidence into the squared-Hodge carrier                                                                   | Exact current-source no-entailment; unrestricted enriched-packet occupation open                            |
| Separator handoff | One coherent $\mathfrak{J}_* = (\Pi_*, \varphi_*)$ fixes the mixed ideal, multiplicity pairing, relative P3–P4 gluing and faithful block occupation         | Conditional joint-source compression; current narrow-source selection and global terminal completeness open |

Table 1: Inherited proof hierarchy. The open top-level source arrow does not erase the lower-level theorems, while the lower-level constructions do not prove unrestricted physical occupation.

This hierarchy fixes the interpretation of the paper. The mathematical P3 and quantum-terminal machinery is already available at several levels. The remaining source question is the

typed incidence

$$\iota_{P3} : (E_{P3}, K_*) \longrightarrow (R_{\text{common}}, h_{\text{common}}), \quad (3)$$

selected before the handoff. Existing EACER support is rank two; the rank-nine seed-normal and endpoint/context carriers lack the required typed map; and the common Clifford envelope is an operator target rather than the source incidence. Paper VI uses the adopted completion for forward terminal theorems and keeps Eq. (3) open at the unrestricted physical parent-law level.

## 2 Question, architecture and claim boundary

Let  $M_\tau$  denote a stabilized atemporal morphological body generated by the physically enriched Tau base and a universal seed. For an observer carrier  $O$ , source region  $S$  and context  $c$ , the accessible terminal information is not the body itself. It is obtained through a body-conditioned relation and a stable finite-resolution map,

$$B_\tau^{\text{phys}} + s_U \longrightarrow M_\tau \longrightarrow \mathcal{C}_{OS|c}(M_\tau) \xrightarrow{Q_{OS|c}} \mathcal{R}_{OS|c}. \quad (4)$$

Here  $\mathcal{C}_{OS|c}$  is not a separate substance. It is the observer–source realization of relations already conditioned by the body. The stable quantizer  $Q_{OS|c}$  defines observer-relative readout classes. The label  $O$  initially denotes a body-side carrier candidate. Operational observerhood and an accessible four-dimensional world arise together only when the same solution has regular local rank four, a stable quantizer and a nonzero record/effect. Hence information loss is not transmission toward a detached observer, and a 4D chart alone is not a measurement context.

Paper VI inherits, rather than proves, the minimal formation factorization. The base is the protected zero-source response structure of one joint law, the universal seed is a response-distinct base-relative source class, and  $M_\tau$  is a stable selected nonneutral response. The split is separately identifiable only after a physical zero-source member is fixed. Quantum effects, Born statistics and process tomography are downstream and cannot define that anchor or construct the seed backward from terminal data. Equality of classes, rather than a non-transitive distance threshold, defines the operational equivalence relation

$$x \sim_{OS|c} y \iff Q_{OS|c}(x) = Q_{OS|c}(y). \quad (5)$$

The central question is finite and deliberately narrow:

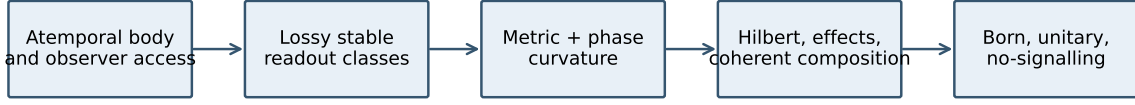
Which additional source-owned structures turn the quotient  $\mathcal{C}_{OS|c}/\sim_{OS|c}$  into a complex quantum carrier, and which familiar quantum laws then follow without further fitted gains?

This co-descent interpretation sharpens the quantum claim boundary. Finite observer-relative quantization belongs to the formation of operational readout classes, but it does not by itself imply complex phase composition, Born probabilities, tensor products, entanglement or no-signalling. Those remain additional source-owned structures derived conditionally below.

Figure 1 displays the complete logical direction. In particular, terminal observations constrain a frozen forward construction; they do not ontologically build  $M_\tau$  backwards.

## 3 Relation to established structures and novelty boundary

The terminal mathematics used below is standard once its inputs are granted. Stinespring/Kraus theory supplies completely positive channels [1, 2]; Lindblad dynamics supplies a standard decoherence limit [3]; Uhlmann geometry supplies mixed-state purification distance [4]; Gleason-type results constrain noncontextual probability measures [5]; and quantum error correction supplies



information loss alone is insufficient

Figure 1: The Paper-VI descent. Finite resolution supplies stable classes, whereas complex phase geometry, effects and composition require additional source-owned structure. Dashed implications are conditional.

exact recoverability criteria [6, 7]. Quantum-geometric response and speed-limit relations are likewise established structures [8, 9]. Paper VI does not claim these results as new consequences of information loss.

The novelty claim is narrower and conditional. Tau Core asks whether one body-derived, observer-relative source packet can own the metric, phase, action unit, effects, subsystem composition and recovery domain together, rather than importing them as unrelated terminal axioms. The paper gives finite sufficient conditions, countermodels to weaker packets and one proposed cross-domain discriminator. It does not prove that the physical base-seed system occupies the sufficient packet. This distinction separates the paper’s source-architecture proposal from its use of standard quantum and QFT theorems.

## 4 The common accessibility quotient

**Definition 1** (Stable observer quantizer). *A stable observer quantizer is an idempotent measurable map  $Q_{OS|c} : \mathcal{C}_{OS|c} \rightarrow \mathcal{R}_{OS|c}$  whose fibres form a partition on the occupied source family and remain invariant under the declared record-tolerance class. Its finite capacity is the number, measure or effective dimension of occupied distinguishable fibres.*

This definition retains observer dependence. Different carriers, source supports or contexts may induce different partitions. Microscopic effective discreteness and macroscopic effective continuity can therefore coexist: a fine carrier resolves fewer equivalence identifications, while redundant compatible records make a dense family of transitions operationally smooth.

For an ordered source path  $\gamma$ , time quantization is the smallest ordered transition whose endpoints occupy distinct stable fibres. It is not a universal atom of time unless independence from  $O, S, c$  is proved. Likewise, multiple parent configurations in one ordinary class are merely unresolved; they are not yet a coherent superposition.

## 5 Local transition resolvability and record-jet detection

The stable quantizer makes the smallest operational transition a derived observer–source scale, not an infinitesimal property of the continuous carrier. Let  $y(\sigma)$  be a source-frozen differentiable local ordered precursor and put

$$v = y'(\sigma_0) \neq 0, \quad d_e(\delta) = y(\sigma_0 + \delta) - y(\sigma_0). \quad (6)$$

Continuity gives, for all sufficiently small  $\delta > 0$ ,

$$\langle d_e(\delta), v \rangle > \frac{\delta}{2} \|v\|^2 > 0, \quad \frac{d_e(\delta)}{\delta} \longrightarrow v. \quad (7)$$

Thus a short positive chord has positive tangent overlap and asymptotically unit alignment. This statement is local: a long curved chord may be orthogonal to its initial tangent.

Let  $\mathcal{R}_{OS}$  be the complete source-frozen record descriptor and let  $C_0$  be the stable  $Q_{OS}$  cell containing  $r_0 = \mathcal{R}_{OS}(0)$ . Assume, in one inherited norm,

$$B_{r_-}(r_0) \subseteq C_0 \subseteq B_{r_+}(r_0), \quad 0 < r_- \leq r_+, \quad (8)$$

and along Eq. (6)

$$\mathcal{R}_{OS}(d_e(\delta)) - r_0 = \delta a + \eta(\delta), \quad a = D\mathcal{R}_{OS}|_0(v), \quad \|\eta(\delta)\| \leq \frac{C}{2}\delta^2. \quad (9)$$

**Proposition 1** (Stable-cell resolution window). *Under Eqs. (8)–(9),*

$$\delta\|a\| + \frac{C}{2}\delta^2 < r_- \implies Q_{OS}(d_e(\delta)) = Q_{OS}(0), \quad (10)$$

$$\delta\|a\| - \frac{C}{2}\delta^2 > r_+ \implies Q_{OS}(d_e(\delta)) \neq Q_{OS}(0). \quad (11)$$

For  $C > 0$  and  $\|a\|^2 > 2Cr_+$ , the smaller sufficient exit scale is

$$\delta_Q^+ = \frac{\|a\| - \sqrt{\|a\|^2 - 2Cr_+}}{C}. \quad (12)$$

If the local geometric result holds to  $\delta_{\text{loc}}$ , a nonempty resolved-local window exists when  $\delta_Q^+ < \delta_{\text{loc}}$  and  $\delta_Q^+ < \delta < \delta_{\text{loc}}$  is chosen on the increasing branch.

*Proof.* The triangle inequalities applied to Eq. (9) give

$$\delta\|a\| - \frac{C}{2}\delta^2 \leq \|\mathcal{R}_{OS}(d_e(\delta)) - r_0\| \leq \delta\|a\| + \frac{C}{2}\delta^2.$$

The inner inclusion in Eq. (8) proves Eq. (10); the outer inclusion proves Eq. (11). Solving equality in the latter lower bound gives Eq. (12).  $\square$

Consequently nonzero continuous change is not yet time or quantum quantization. The physical chain must source both the record jet and the stable partition,

$$(K_{\text{src}}, J_{OS}^{\text{acc}}) \longrightarrow P_O \longrightarrow \{\mathcal{I}_a\}_a \longrightarrow \mathcal{R}_{OS} \longrightarrow Q_{OS}^{\mathcal{I}}. \quad (13)$$

At constant rank define

$$B(M) = J_{OS}^{\text{acc}}(M)K_{\text{src}}^{-1/2}, \quad P_O = B^\dagger(BB^\dagger)^+B, \quad W = B^\dagger. \quad (14)$$

Its Kato projector jet in direction  $v$  is

$$D_v P_O = X_v + X_v^\dagger, \quad X_v = (I - P_O)(D_v W)W^+. \quad (15)$$

An internal basis change of the access image can have  $D_v W \neq 0$  but  $X_v = 0$ ; base incidence and its metric therefore do not determine outward motion. If  $W = SR$ , the correctly typed nonstandard candidate is the standard-access-quotiented covariant second fundamental form

$$\Pi_{OS}^\perp(v) = (I - \Pi_{\text{std}}^{\text{acc}})(I - P_O)S\nabla_v^{(K)}R. \quad (16)$$

Its nonvanishing guarantees nonstandard outward access motion, not by itself finite-cell exit.

**Theorem 1** (Exhaustive complete-Choi detection). *Let  $X_v \neq 0$ ,  $G_v = X_v - X_v^\dagger$ , and let an owned finite instrument have effect roots  $S_a = \sqrt{F_a}$  exhaustive on the accessible support,*

$$\sum_a F_a = \sum_a S_a^2 = P_O. \quad (17)$$

*Under Kato transport,  $\dot{K}_a = G_v S_a$  and  $J_a = |K_a\rangle\langle K_a|$ . Then*

$$X_v \neq 0 \implies \dot{J}_a \neq 0 \text{ for at least one } a. \quad (18)$$

*The immediate effects may nevertheless obey  $\dot{F}_a = 0$ .*

*Proof.* The ranges of  $S_a$  and  $G_v S_a = X_v S_a$  are orthogonal, so  $\dot{J}_a = 0$  exactly when  $X_v S_a = 0$ . If all Choi derivatives vanished, exhaustivity would give

$$X_v = X_v P_O = X_v \sum_a S_a^2 = \sum_a (X_v S_a) S_a = 0,$$

a contradiction. Orthogonality also gives  $\dot{F}_a = (G_v S_a)^\dagger S_a + S_a G_v S_a = 0$ .  $\square$

**Corollary 1** (Full-support and scalar-phase controls). *If the direct access realization has  $P_O = I_n$ , then*

$$X_v = (I - P_O)(D_v W)W^+ = 0, \quad G_v = X_v - X_v^\dagger = 0, \quad (19)$$

*so the Kato branch contains no outward access transport even when the terminal process varies. If instead a Kraus amplitude is transported only by a scalar generator  $G_v = ia(v)I_n$ , its normalized Choi projector is unchanged by the resulting global phase and*

$$q_{\text{pu}}(G_v) := G_v - \frac{\text{tr } G_v}{n} I_n = 0. \quad (20)$$

*Therefore a non-scalar normalized-Choi orientation requires either the internal component of a fuller body connection or a genuinely varying access projector.*

*Proof.* The first statement follows directly from Eq. (15). For the second, scalar transport multiplies every Kraus vector by one phase; the outer product  $|K\rangle\langle K|$  is phase invariant, and the displayed projective quotient vanishes identically.  $\square$

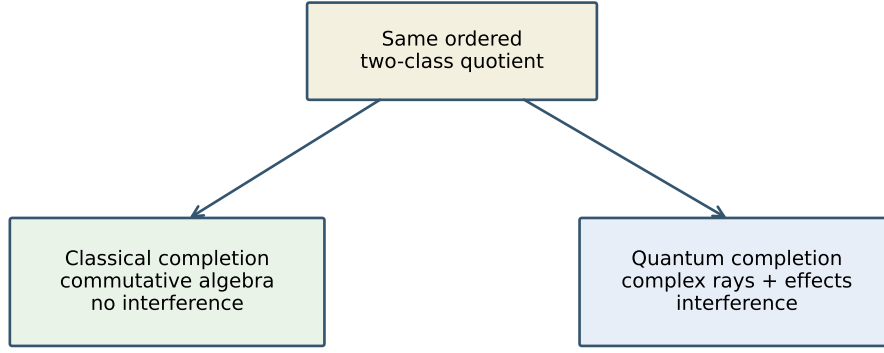
Probability-only observers and incomplete Choi families can therefore miss an outward access jet. Conversely, exhaustive detection does not make the signal Tau-specific: the additional discriminator is a nonzero component after projection away from independently frozen standard apparatus Choi tangents. Even that infinitesimal component must still satisfy Proposition 1 to become a finite resolved record. Physical joint occupation of the front, incidence, complete instrument and stable cell remains open.

## 6 Loss-only no-go

**Proposition 2** (Loss-only no-go). *Finite capacity, non-injectivity and a stable ordered quotient do not imply a complex Hilbert space, phase-sensitive composition, the Born rule, entanglement or no-signalling.*

*Proof.* Take the reported state space to be the classical bit simplex  $\Delta_2$ . One completion is  $\Delta_2$  itself with stochastic maps. A second completion is the qubit density-operator space followed by complete dephasing in a fixed basis,

$$\mathcal{D}(\rho) = \text{diag}(\rho) \in \Delta_2. \quad (21)$$



quotient and finite capacity do not select the branch

Figure 2: The same finite observer quotient admits a classical simplex and a complex quantum carrier. The distinguishing datum is phase-capable coherent composition, not information loss by itself.

Both have exactly the same resolved quotient, occupied probability states and finite outcome alphabet. Before dephasing, the quantum completion also has off-diagonal phase and interference, whereas the classical completion does not. Adding the same order to the two reported outcomes changes neither construction. Hence the common loss/quotient data cannot select any of the listed quantum properties.  $\square$

This no-go protects the program from an otherwise tempting circularity. It also explains why observer-dependent quantization is necessary but not sufficient. Figure 2 summarizes the fork.

## 7 Frozen quantum source packet

We now state the smallest packet used in the positive construction. Every item must arise from one frozen post-body source; the packet may not be assembled by fitting terminal residuals.

**Definition 2** (Quantum source packet). *On an occupied finite-dimensional horizontal tangent space  $V_O$ , let*

$$\mathfrak{P}_Q = (g, \theta, \omega, U_A, K, \mathfrak{E}, \otimes), \quad \omega = d\theta, \quad (22)$$

where:

- (Q1)  $g$  is the positive quotient of the body Hessian;
- (Q2)  $\theta$  is a source-owned ordered-action one-form and  $\omega$  its nondegenerate curvature on the quantum leaf;
- (Q3)  $A = g^{-1}\omega$  obeys  $A^2 = -\kappa^2 I$  for one  $\kappa > 0$ ;
- (Q4)  $U_A > 0$  is the common action unit, with no block-dependent gain;
- (Q5)  $K$  is  $g$ -self-adjoint and commutes with  $J = A/\kappa$ ;
- (Q6)  $\mathfrak{E}$  is a normalized positive effect family represented by the source-owned trace state;
- (Q7)  $\otimes$  is a declared composition law on independently accessible occupied subsystems, and local deterministic operations are CPTP.

Items (Q1)–(Q3) are the metric–symplectic completion. Items (Q4)–(Q6) select dynamics and readout statistics. Item (Q7) is logically independent: no-signalling cannot be proved before a subsystem composition is defined.

## 8 Existing Tau source realization

The packet can now be evaluated against earlier Tau source results rather than left as a free list. Inside the declared source-irredundant enriched MVP, the existing occupation theorem supplies an occupied coherent P3 representation and a faithful record edge. The adopted graph-normal P3 handoff has stationary operator coordinate

$$S_{P3} = c_K(e_{P3}) \quad (23)$$

in the complex Clifford carrier, with one action unit  $\mathcal{A}_*$  and the normalized trace. Its finite source-minimal block ledger is

$$\text{Cl}_7(\mathbb{C}) \simeq M_8(\mathbb{C}) \oplus M_8(\mathbb{C}). \quad (24)$$

Restrict to one occupied simple block  $\mathcal{A}_Q = M_n(\mathbb{C})$  and let  $\tau_n = \text{Tr}/n$ . On  $\mathcal{A}_Q$  define

$$h(X, Y) = \mathcal{A}_* \tau_n(X^\dagger Y), \quad g = \text{Re } h, \quad JX = iX, \quad \omega(X, Y) = g(JX, Y). \quad (25)$$

Because the matrix block is an affine vector space, the canonical linear potential

$$\theta_X(Y) = \frac{1}{2} \omega(X, Y) \quad (26)$$

satisfies  $d\theta = \omega$ . Thus the construction supplies the one-form in (Q2), not only its curvature.

**Theorem 2** (Existing-source single-system packet). *Within the adopted enriched MVP/P3 hand-off, Eqs. (25)–(26) construct (Q1)–(Q4) on each occupied simple block in one source action unit. Every handoff-selected Hermitian generator and positive normalized effect family is then represented in the canonical forms required by (Q5)–(Q6). No independent metric, phase or Born gain is required; the physical choice of generator and occupied effects remains source data.*

*Proof.* Faithfulness of  $\tau_n$  makes  $g$  positive. Scalar multiplication gives  $J^2 = -I$  and preserves  $g$ , so  $A = g^{-1}\omega = J$  and (Q3) holds with  $\kappa = 1$ . Antisymmetry of  $\omega$  follows from Hermitian symmetry, Eq. (26) supplies  $d\theta = \omega$ , and nondegeneracy follows by choosing  $Y = JX$ :  $\omega(X, JX) = g(X, X)$ . For  $H = H^\dagger$ , left multiplication  $K_H(X) = HX$  is self-adjoint for  $h$  and commutes with  $J$ , so  $\exp(tJK_H/\mathcal{A}_*)$  is unitary. Positive matrices  $r, E$  with  $\tau_n(r) = 1$  and  $0 \leq E \leq I$  define the standard terminal density  $\rho = r/n$ , for which

$$\text{Tr } \rho = 1, \quad p(E|\rho) = \tau_n(rE) = \text{Tr}(\rho E). \quad (27)$$

Positivity, normalization and coherent interference follow from this same pairing. All structures are restrictions of the trace-normalized handoff.  $\square$

This consolidates earlier results but has two sharp boundaries. First, the unchanged explicit parent does not select the required full-role squared-Hodge incidence: its retained phase carrier is only  $\mathbb{CP}^0$  and its phase/body mixing block is exactly zero. The theorem therefore remains conditional on the already adopted P3 handoff. Second,  $M_8(\mathbb{C})$  is abstractly isomorphic to  $M_2(\mathbb{C})^{\otimes 3}$ , but the simple algebra and trace do not select a unique tensor factorization. Inner automorphisms preserve the single-system packet while moving candidate local factors. Thus (Q7), entanglement locality and subsystem no-signalling still require a source-owned commuting-factor ledger.

The remaining handoff condition has a scalar record formulation. Let  $C : E_{P3} \rightarrow \mathcal{A}_{\text{occ}}^{\text{sa}}$  be a linear typed response and let  $\Phi_{\text{sep}}$  be a source-owned separating family of normalized positive functionals.

**Theorem 3** (Separating-record selector). *If, for every  $v$  and every  $\varphi \in \Phi_{\text{sep}}$ ,*

$$\varphi(C(v)^2) = K_*(v, v), \quad (28)$$

*then  $C(v)^2 = K_*(v, v)I$ , polarization gives the Clifford relations, and the strictly convex complete-mediation theorem selects the graph-normal handoff without a bypass or new gain.*

*Proof.* The self-adjoint difference  $D_v = C(v)^2 - K_*(v, v)I$  is annihilated by every member of a separating family, so  $D_v = 0$ . Applying this equality to  $u + v$  and subtracting the diagonal instances gives  $C(u)C(v) + C(v)C(u) = 2K_*(u, v)I$ . Strict convexity then makes  $S_{P3} = C(e_{P3})$  the unique mediated zero locus.  $\square$

This reduces the unrestricted parent-handoff question to a finite scalar source law, but does not assume it has been observed. Normalized-trace agreement alone is insufficient:  $D(v) = \sqrt{2} \text{diag}(v_1, v_2)$  has the correct averaged quadratic trace, while its two pure diagonal records return different squares. Paper III supplies an operational informational-completeness criterion for  $\Phi_{\text{sep}}$ ; it does not prove physical occupation of Eq. (28).

### 8.1 Finite P3 separating-record rank

The separating requirement can be evaluated without introducing another operator family. On one irreducible P3 block let  $\gamma_1, \dots, \gamma_7$  be Hermitian Clifford generators and, for an ordered subset  $A = \{a_1 < \dots < a_k\}$ , set

$$W_A = i^{k(k-1)/2} \gamma_{a_1} \dots \gamma_{a_k}. \quad (29)$$

Each  $W_A$  is Hermitian, so  $\rho \mapsto \text{Tr}(\rho W_A)$  is a real record coordinate.

**Theorem 4** (P3 separating-record rank). *On one irreducible  $M_8(\mathbb{C})$  block, the real ranks of the cumulative record families with Clifford grades at most 0, 1, 2, 3 are*

$$\boxed{1, \quad 8, \quad 29, \quad 64}, \quad (30)$$

*and their unresolved self-adjoint kernel dimensions are respectively*

$$\boxed{63, \quad 56, \quad 35, \quad 0}. \quad (31)$$

*Consequently, the unit and seven primitive P3 records are not separating, and even complete binary-word occupation is insufficient. Complete occupation through ternary words is separating.*

*Proof.* The grade multiplicities are  $\binom{7}{0} = 1$ ,  $\binom{7}{1} = 7$ ,  $\binom{7}{2} = 21$  and  $\binom{7}{3} = 35$ . In an irreducible odd-Clifford block the volume word is scalar and Hodge duality identifies grades  $7 - k$  and  $k$ . The words through grade three are orthogonal under normalized trace and total  $1 + 7 + 21 + 35 = 64 = \dim_{\mathbb{R}} M_8(\mathbb{C})_{\text{sa}}$ ; hence they form a real basis. Truncation gives Eq. (30), and subtraction from 64 gives Eq. (31).  $\square$

This closes the rank question but not the source question. Algebraic generation of  $W_A$  does not make it a physically occupied stable record, and Frobenius copying of an occupied record does not make the record family informationally complete. The remaining decision is finite: whether the base-seed record law occupies all  $|A| \leq 3$  words with the common scalar action in Eq. (28). No grade-four-or-higher record is needed on this block.

### 8.2 Canonical spectral occupation and the Frobenius no-go

The rank theorem also identifies a canonical positive record realization. Every Hermitian Clifford word is an involution, so define

$$E_A^{\pm} = \frac{1}{2}(I \pm W_A). \quad (32)$$

These are orthogonal positive effects and  $E_A^+ + E_A^- = I$ . Different noncommuting words label different binary contexts, not one joint pointer partition.

**Theorem 5** (Ternary spectral-record occupation). *If the physical instrument is spectrally closed on every occupied grade-at-most-three Clifford involution, with the common trace/effect pairing and no context-dependent gain, the resulting 63 nontrivial binary instruments form a separating positive record atlas. Together with Eq. (28), this closes the record hypothesis of Theorem 3.*

*Proof.* For every state  $\rho$ ,

$$p_A^+ - p_A^- = \text{Tr } \rho(E_A^+ - E_A^-) = \text{Tr}(\rho W_A). \quad (33)$$

The words through grade three form a real basis by Theorem 4. Equal binary outcome probabilities therefore imply equal expectations on a basis and hence equal states.  $\square$

This spectral closure is not a consequence of the current Frobenius record law. A countermodel retains the same matrix algebra and all Clifford words, but makes only the eight projectors of one distinguished diagonal basis into physical pointer records. The coproduct  $|k\rangle \mapsto |k\rangle \otimes |k\rangle$  remains exact, special and coassociative, while the record span has rank only eight and misses every off-diagonal state direction. Thus redundant pointer copying and algebraic word generation do not imply complete spectral-record occupation. The remaining source law is finite and explicit:

$$W = W^\dagger, \quad W^2 = I, \quad |A| \leq 3 \implies \{(I + W)/2, (I - W)/2\} \subset \mathcal{E}_{\text{rec}}^{\text{occ}}. \quad (34)$$

Its selection by the physical base-seed instrument action remains open.

The previously established effect/instrument branch closes this implication inside the adopted enriched MVP. If S4 is the full effect interval  $\{E \in \mathcal{A}_{\text{occ}}^{\text{sa}} : 0 \leq E \leq I\}$  and S5 assigns the apparatus-free square-root instrument to every sourced effect, then Eq. (32) already belongs to S4 and

$$\mathcal{I}_A^\pm(\rho) = E_A^\pm \rho E_A^\pm \quad (35)$$

is its sharp S5 terminal. The two maps have a trace-preserving sum and each selected outcome is immediately repeatable. Therefore

P3HAND-D1 + full S4 effect interval + S5 terminal  $\implies$  ternary spectral-record occupation.

(36)

No further instrument gain or apparatus orientation is needed. This is an existing-source theorem inside the full enriched packet, not a derivation from the unchanged narrow parent: unrestricted realization and occupation of P3HAND-D1 and the complete S4/S5 packet remain open.

There is a strict directionality boundary. This argument assumes P3HAND-D1 to obtain the involutions  $W_A$  and therefore cannot be used backwards in Theorem 3 to select that same handoff. It closes the record atlas after adoption. A noncircular P3 selector would require the separating scalar evaluations on a pre-handoff source carrier defined without the Clifford representation.

## 9 Metric–symplectic quantum completion

**Theorem 6** (Metric–symplectic quantum completion). *Under (Q1)–(Q3),  $J = A/\kappa$  is an orthogonal complex structure. The pair  $(g, J)$  defines a Hermitian form*

$$h(u, v) = g(u, v) - ig(Ju, v), \quad (37)$$

*and therefore a finite complex Hilbert carrier after completion and quotient by any common null directions.*

*Proof.* Because  $\omega$  is antisymmetric,  $g(Au, v) = -g(u, Av)$ . Thus  $A$  is  $g$ -skew-adjoint. The lock gives  $J^2 = -I$ . Moreover

$$g(Ju, Jv) = -g(u, J^2v) = g(u, v), \quad (38)$$

so  $J$  is  $g$ -orthogonal. Equation (37) is conjugate symmetric, complex-linear under multiplication by  $J$ , and positive because  $h(u, u) = g(u, u) > 0$ . Hence it is a Hermitian inner product on the physical quotient.  $\square$

**Remark 1.** *The lock is a physical completion condition, not a theorem of finite loss. If  $A^2$  has unequal negative eigenvalues, one obtains mode-dependent complex scales; if  $A$  is degenerate, part of the carrier is phase-blind. Both are testable alternative branches and neither may be silently normalized away.*

## 10 Unitary transport and the terminal clock

Paper V supplies an observer-relative calibrated duration  $t_O$  on a regular ordered leaf. It does not by itself supply quantum dynamics. Under (Q4) and (Q5), define

$$X = JK/U_A. \quad (39)$$

Since  $K$  is  $g$ -self-adjoint and  $[K, J] = 0$ ,  $X$  is  $g$ -skew-adjoint and complex-linear. Therefore

$$U(t_O) = e^{t_O X} \quad (40)$$

preserves  $h$ . In complex notation this is

$$iU_A \frac{d}{dt_O} |\psi\rangle = H |\psi\rangle, \quad H = H^\dagger. \quad (41)$$

The identification  $U_A = \hbar$  is fixed by the standard quantum limit. Paper VI explains the role of the constant: it is the common action conversion between ordered phase transport and calibrated terminal time. It does not derive the numerical SI value of  $\hbar$  from the minimal Tau parent.

## 11 Born and interference terminal

**Theorem 7** (Born and interference terminal). *Let  $\rho \geq 0$ ,  $\text{Tr } \rho = 1$  be an occupied carrier state and let  $\{E_a\}$  be positive effects with  $\sum_a E_a = I$ . If the source-owned record functional is the normalized positive trace pairing, then*

$$p(a|\rho) = \text{Tr}(\rho E_a) \quad (42)$$

*is a normalized probability law. For a rank-one record and coherent alternatives  $\psi = \psi_1 + e^{i\phi}\psi_2$ , it contains the cross term*

$$\|\psi\|^2 = \|\psi_1\|^2 + \|\psi_2\|^2 + 2 \text{Re}\left(e^{i\phi} \langle \psi_1, \psi_2 \rangle\right). \quad (43)$$

*Proof.* Positivity gives  $p(a|\rho) \geq 0$ , while normalization gives  $\sum_a p(a|\rho) = \text{Tr } \rho = 1$ . The second statement follows by expanding the Hermitian norm. The phase-sensitive cross term is absent from a classical mixture, which instead has  $\rho = \rho_1 + \rho_2$  without off-diagonal coherence.  $\square$

In dimensions at least three, noncontextual frame measures are constrained by Gleason-type representation results [5]. We do not use this as a derivation of (Q6): the physical realization of noncontextual positive effects and the source trace remains part of the joint packet. This avoids claiming the Born rule from Hilbert-space notation alone.

## 12 Record transport and recoverability

The finite Uhlmann geometry developed in Paper III enters here only as a transport condition. A noisy record edge is a CPTP map  $\mathcal{C}$ . On a declared occupied operator system  $\mathcal{S}_{\text{occ}}$ , exact recovery means there exists CPTP  $\mathcal{R}$  such that

$$(\mathcal{R} \circ \mathcal{C})(X) = X, \quad X \in \mathcal{S}_{\text{occ}}. \quad (44)$$

Then the distinguishability used by the terminal is preserved on that domain; outside it, contractivity is the correct generic statement. Uhlmann amplitudes provide the purification geometry [4], while Stinespring dilation and Kraus representations supply the standard channel framework [1, 2].

Exact and approximate recoverability, information–disturbance and operator-system correction are standard quantum-information structures; they are not claimed as Tau innovations. Their role here is diagnostic: they state what the observer descent must preserve if the same body-derived packet is to support a complete logical quantum terminal. The specifically Tau-dependent content is the proposed common source of that recoverable operator system and the body-action/Uhlmann identification, neither of which follows from channel theory alone.

This paper distinguishes four domains that are often conflated: a state family, a linear operator system, a code subspace and a correctable algebra. Exact recovery of one does not automatically imply recovery of the full matrix algebra. Approximate recovery requires an explicit error ledger, for example  $\|\mathcal{RC} - \text{id}\|_\diamond \leq \eta$ ; no exact finite-action identity is claimed from this inequality alone.

### 13 Subsystem-factorization boundary

The occupied P3 block is a full matrix algebra  $\mathcal{A}_Q \simeq M_8(\mathbb{C})$ , hence it admits abstract realizations as  $M_2(\mathbb{C})^{\otimes 3}$ . This dimension statement does not identify which observables belong to which physical subsystem. The full S4 effect interval and S5 spectral instruments are likewise invariant under inner automorphisms and therefore do not by themselves break this ambiguity. Accordingly, the tensor-product algebra below is a finite operator-algebraic selector relative to declared source subalgebras, not a derivation of spatial subsystems from matrix dimension alone.

**Proposition 3** (One-record factorization no-go). *Fixing one pointer algebra, or even one factor  $\mathcal{A}_1 \simeq M_2(\mathbb{C})$  inside  $M_8(\mathbb{C})$ , does not select a three-factor tensor structure. The commutant of  $\mathcal{A}_1$  is  $M_4(\mathbb{C})$  and retains multiple candidate decompositions into two  $M_2(\mathbb{C})$  factors relative to the frozen source descriptor.*

*Proof.* In the standard representation let  $\mathcal{A}_1 = M_2 \otimes I_2 \otimes I_2$ . Its commutant is  $I_2 \otimes M_4$ . Conjugation inside this commutant moves candidate second and third factors while fixing  $\mathcal{A}_1$ . More strongly, the diagonal entangling family  $U_\theta = \exp(i\theta Z \otimes Z \otimes I)$  fixes every computational-basis pointer projector pointwise while conjugating the first two local Pauli algebras out of their original spans for generic  $\theta$ . It preserves the normalized trace, the full effect interval and joint generation of  $M_8$ . Thus neither one record atlas nor one selected factor removes the ambiguity.  $\square$

**Theorem 8** (Two-factor selector). *Suppose the source owns two commuting unital subalgebras of  $M_8(\mathbb{C})$ , denoted by  $\mathcal{A}_1$  and  $\mathcal{A}_2$ . Each is isomorphic to  $M_2(\mathbb{C})$ , and their multiplication representation is faithful:*

$$C^*(\mathcal{A}_1, \mathcal{A}_2) \simeq M_4(\mathbb{C}). \quad (45)$$

Then

$$\mathcal{A}_3 = C^*(\mathcal{A}_1, \mathcal{A}_2)' \simeq M_2(\mathbb{C}) \quad (46)$$

and the three factors jointly generate  $M_8(\mathbb{C})$ . The resulting factorization is unique relative to the selected pair, up to local inner automorphisms and permutation of equally typed factors.

*Proof.* Finite-dimensional representation theory puts the faithful commuting pair in the form  $M_2 \otimes M_2 \otimes I_2$ . The double-commutant theorem gives Eq. (46); multiplying by its commutant produces  $M_2 \otimes M_2 \otimes M_2 = M_8$ . Skolem–Noether leaves only inner changes inside the selected simple factors, together with permutations allowed by their source types.  $\square$

The finite audit gives factor dimensions (4, 4, 4), joint complex operator rank 64, and commutant complex dimensions (16, 4, 1) after selecting one, two and three factors. The present ROOT edge can anchor a record relation, but the current source ledger does not yet provide the required

commuting  $M_2$  pair. Consequently this theorem localizes rather than closes the physical composition problem. The P3 roles, ROOT edge and complete S4/S5 atlas construct a single-system quantum packet but do not yet derive subsystem locality.

There is a more source-proximal form of the selector. Let an *oriented P3 Clifford flag* be an orthogonal decomposition of the seven-dimensional P3 role space into three oriented two-planes and one oriented line. For an adapted Clifford frame  $\gamma_1, \dots, \gamma_7$ , put

$$z_j = -i\gamma_{2j-1}\gamma_{2j}, \quad x_j = \left( \prod_{k < j} z_k \right) \gamma_{2j-1}, \quad y_j = \left( \prod_{k < j} z_k \right) \gamma_{2j}. \quad (47)$$

The Clifford relations imply that  $(x_j, y_j, z_j)$  generates one  $M_2$  factor, different triples commute, and their product generates  $M_8$ . Therefore a source-owned Clifford flag is sufficient to realize the two-factor selector and its third commutant. The present isotropic P3 metric, typed role inventory and one occupied ROOT edge do not select such a flag. Before orientation and factor ordering there are already  $7(6-1)!! = 105$  choices of a residual line and a pairing of the other six axes. A complete ROOT-edge matching or another independently required source polarization could select one; mere role naming cannot.

The tetrahedral seed gives a tempting but incomplete realization. Its six edge labels possess the canonical partition into three opposite-edge pairs, so they reproduce the combinatorial shape of the required flag. However, the existing equivariant CHDF pair map from four seed roles into those six labels has rank four. Its two-dimensional [22] complement is precisely the sector annihilated by the current scalar-plus-quadrupole curvature map, and the root-stabilizer supplies no nonzero direct load into it. Therefore the present tetrahedral body cannot provide six independent metric directions and cannot realize the Clifford flag. This is a rank obstruction, not a failure of the opposite-edge pairing idea. The flag must instead come from a richer base-seed response that physically activates the missing doublet, or from an independently source-owned post-body P3 polarization.

This also joins the factorization problem to the central-block problem. In an irreducible  $\text{Cl}_7(\mathbb{C})$  block the seventh Clifford generator is, up to the chosen chirality convention, the product  $z_1 z_2 z_3$ . Thus a full six-edge flag together with one source-owned chirality sign selects both an  $M_8$  summand and its three  $M_2$  factors. A minimal positive completion can retain the established rank-four CHDF equilibrium while realizing the missing [22] doublet as a zero-background, positive-Hessian fluctuation sector. It does not require a nonzero symmetry-breaking load. What remains unproved is that the physical base assigns this doublet the same edge action unit and selects one chirality. These are two clauses of one finite packet, not new terminal choices.

**Proposition 4** (Tetrahedral-packet independence). *The current base-seed axioms do not entail either positive [22] stiffness or a  $\text{Cl}_7$  central character. Consequently they do not entail the six-edge/chirality tensor packet.*

*Proof.* On the established rank-four CHDF image, append a [22] coordinate  $w$  with quadratic term  $\lambda \langle w, w \rangle / 2$ . Both  $\lambda = 0$  and every  $\lambda > 0$  preserve the same equilibrium  $w = 0$ , the same CHDF loading and all current  $S_4$  covariance data, while only the latter supplies a positive two-dimensional fluctuation metric. Hence current data do not force  $\lambda > 0$ . Independently, the two irreducible  $M_8$  representations differ by the sign of the central Clifford volume element. The occupied seed sign  $\nu = +1$  exists before this odd Clifford element is introduced and is not a character of its center; both central signs are compatible with the current seed data. Thus neither required clause is entailed.  $\square$

Conversely, a common-unit term  $\mathcal{A}_* \langle w, w \rangle / 2$  together with a source character  $\chi(\Omega) = \epsilon \in \{\pm 1\}$  is sufficient. The first completes the six-edge metric, the opposite-edge matching supplies the three planes, and the second chooses the simple block and fixes  $\gamma_7 = \epsilon z_1 z_2 z_3$ . Equation (47) then constructs the tensor factors. This is a finite sufficient common-source completion relative to the

frozen tetrahedral matching. A positive [22] metric and a central character are necessary data types for this route, but factorization alone would not fix the metric coefficient to  $\mathcal{A}_*$ . Physical occupation of both clauses remains open.

This enrichment is deliberately narrow. The tetrahedral edge representation decomposes as  $[4] \oplus [31] \oplus [22]$ , while the established CHDF image is exactly  $[4] \oplus [31]$ . Hence [22] is the unique missing source type and its dimension is exactly the rank deficit. Moreover, Schur's lemma makes the space of  $S_4$ -invariant symmetric Hessians on this irreducible doublet one-dimensional:

$$K_{W_2} = \lambda_{W_2} I_{W_2}. \quad (48)$$

The audit verifies this invariant-Hessian dimension directly over all 24 tetrahedral permutations. Thus the proposed body enrichment introduces no arbitrary tensor or response function. It remains inadmissible as physical input until the parent fixes  $\lambda_{W_2} > 0$  and its action unit without terminal fitting. The  $\lambda_{W_2} = 0$  model is the preregistered rejection control.

There is a source-derived route that avoids adjoining an independent doublet. The tetrahedral decoder identifies the six-edge carrier with one scalar plus the five-dimensional STF representation. The compact seed supplies a rank-three STF orbit, while source-frozen base occurrence transports generate

$$V_{\text{gen}} = \text{span}_p \rho_2(P_p) W_{\text{seed}}. \quad (49)$$

The existing covariant body solve is positive on this generated carrier, and full edge completion is equivalent to  $\dim V_{\text{gen}} = 5$ . An explicit two-frame witness reaches rank five, so the missing doublet can arise from the same seed realized in distinct base-owned relational frames, without new seed payload or a quantum-terminal fit. Physical closure still requires the actual base holonomy to have rank five and the inherited edge Gram to make the three opposite-edge planes mutually orthogonal. Thus occurrence transport derives a conditional  $W_2$  source, not yet the complete Clifford flag.

The remaining Gram test is negative for the naive opposite-edge flag. Under the natural Frobenius metric, opposite edges have zero inner product within a pair, but every entry between two distinct pair planes is  $1/4$ . The edge-Gram spectrum is  $(1/2, 1/2, 1, 1, 1, 2)$ , and pair-common/difference coordinates give the tetrahedral decomposition  $([4] \oplus [22]) \oplus [31]$ , not three orthogonal two-planes. Thus occurrence transport can source the full carrier and its positive Hessian without selecting quantum subsystems. A further source-owned polarization must pair the three relative directions with an orthonormal frame of the scalar-plus-doublet sector; it cannot be chosen by terminal scoring.

A physical root makes this final polarization type-correct. Restriction from  $S_4$  to the root stabilizer  $S_3$  sends both three-dimensional sides to  $\mathbf{1} \oplus W_2$ . Schur's lemma therefore gives every equivariant mixed Gram in the finite form

$$C_{\text{root}} = c_1 P_1 + c_2 P_{W_2}. \quad (50)$$

If  $c_1 c_2 \neq 0$ , its polar part is the unique metric-compatible rooted isometry and selects the three two-planes without separate gains. Root covariance alone permits either coefficient to vanish and permits relative sign controls, so it does not prove the selector. The remaining source audit is exactly whether the frozen body Hessian owns the nondegenerate mixed block in Eq. (50).

**Proposition 5** (Current-body rooted-polarization no-go). *The frozen rank-four CHDF body Hessian cannot own a nondegenerate mixed Gram of the form in Eq. (50). More precisely, its represented pair image contains the complete pair-difference module [31] but only the scalar line of the pair-common module  $[4] \oplus [22]$ . Consequently its rooted common/difference mixed block has rank at most one and necessarily  $c_2 = 0$ .*

*Proof.* The exact pair decomposition is  $\mathbb{R}_{\text{edge}}^6 = [4] \oplus [31] \oplus [22]$ , whereas the current CHDF image is  $\text{Im } \Phi_{\text{sum}} = [4] \oplus [31]$ . Pair differences are precisely the three-dimensional [31] summand and pair commons are  $[4] \oplus [22]$ . Hence the intersections of the current image with the difference

and common subspaces have dimensions three and one, respectively; the common [22] doublet is absent. After restriction to the rooted  $S_3$ , the difference side is  $\mathbf{1} \oplus W_2$ , but the represented common side contains only  $\mathbf{1}$ . Schur equivariance therefore leaves at most the scalar-to-scalar coefficient  $c_1$  and forces  $c_2 = 0$ . The numerical audit checks the intersection dimensions directly. The known nonzero CHDF common/root-standard entry  $-\sqrt{5}/8$  lies in this retained scalar block and does not restore the missing  $W_2$  leg.  $\square$

Occurrence transport can make the missing [22] carrier available, but rank five alone does not create its mixed Hessian with the rooted  $W_2$  inside [31]. Moreover, an unrooted  $S_4$ -equivariant Hessian cannot mix the inequivalent irreducibles [31] and [22]. The route can therefore reopen only through a source-owned rooted mixed incidence whose two  $W_2$  legs, common action unit, adjoint and nondegeneracy descend from the same body functional. This is a localized source obligation, not permission to append another terminal-selected body module.

The hub already contains a conditional realization of precisely this obligation. A source-owned tetrahedral root orders every opposite pair as root-incident/opposite. On the six-edge carrier define the root involution

$$R_* = \text{diag}(1, -1, 1, -1, 1, -1). \quad (51)$$

It exchanges the orthonormal pair-difference and pair-common frames and commutes with the root stabilizer  $S_3$ . Hence its restriction  $J_{\text{root}} : V_{\text{diff}} \rightarrow V_{\text{common}}$  is a canonical rooted isometry, not an independently chosen connector.

**Proposition 6** (Conditional common-potential closure). *Assume that occurrence completion supplies the full six-edge carrier, the source owns the root in Eq. (51), and the adopted TC-CVP1 common strictly convex parent potential restricts to the typed common/difference pair. Then its normalized quadratic germ is*

$$\mathcal{A}_{\text{root}} = \frac{\mu_0}{2} \|c - J_{\text{root}} d\|^2, \quad \mu_0 > 0, \quad (52)$$

and the mixed Hessian is  $H_{cd} = -\mu_0 J_{\text{root}}$ . Consequently  $c_1 = c_2 = -\mu_0 \neq 0$ , so its polar part selects the required rooted tensor polarization without a block-dependent gain.

*Proof.* In every opposite pair, multiplication by  $(1, -1)$  sends the normalized difference vector  $(1, -1)/\sqrt{2}$  to the normalized common vector  $(1, 1)/\sqrt{2}$ . Permutations fixing the root permute the three pairs and therefore commute with  $R_*$ . Thus  $J_{\text{root}}$  is orthogonal and  $S_3$ -equivariant on the complete three-dimensional modules. Differentiating Eq. (52) gives unit pure blocks and the stated mixed block. Its restriction to both irreducible rooted summands is the same nonzero scalar multiple of the identity. The mixed block therefore has rank three and its polar factor is  $-J_{\text{root}}$ .  $\square$

This result reuses the existing rooted-occurrence and common-convex-potential branches; it does not prove their physical joint occupation. The current rank-four CHDF Hessian still obeys Proposition 5. The occurrence results sharpen this boundary: MBC-P1/ACT-M1 already supplies the universal four-certificate support and its source root in the adopted working completion. If those certificates are physical body certificates, TC-CVP1 uniquely supplies the typed joint quadratic germ. Thus root ownership and the mixed block are not two further independent selectors.

**Proposition 7** (Common-source reduction and rank-five independence). *Within the adopted MBC-P1/ACT-M1 plus TC-CVP1 completion, tensor-polarization closure is equivalent, apart from central chirality, to*

$$\dim \text{span}_p \rho_2(P_p) W_{\text{seed}} = 5, \quad (53)$$

where the  $P_p$  are source-frozen occurrence-frame transports. MBC root ownership and the typed common-potential law do not by themselves imply Eq. (53).

*Proof.* MBC-P1/ACT-M1 identifies every occurrence with a source-basic embedding of the same four-certificate cell and selects its pointed root. The physical certificate semantics of TC-CVP1 then gives  $H_{cd} = -\mu_0 J_{\text{root}}$  whenever the complete pair carrier is present, so no additional connector or gain remains. The compact tetrahedral seed orbit spans a rank-three subspace  $W_{\text{seed}} \subset \text{STF}_5$ . If all  $P_p$  preserve this subspace, including the aligned choice  $P_p = I$ , the generated span still has dimension three while the same root, support, unit and TC-CVP1 law remain available. A generic second source-frozen frame can instead enlarge the span to dimension five. These two realizations are an exact equal-root/equal-law counterpart, proving independence. Once the rank is five, Proposition 6 makes the mixed block nondegenerate, proving the stated equivalence inside the adopted completion.  $\square$

Unrestricted physical closure therefore requires one remaining common-occupation fact: the physical base–seed occurrence connection must satisfy Eq. (53). The independently required central chirality is still not selected. The split-potential control has the same pure blocks and zero mixed Hessian, so physical realization remains falsifiable rather than definitional.

The minimum physical completion can nevertheless be closed without another free coefficient. Adopt the following recovery requirement:

**OHC-P1 (irreducible occurrence-holonomy completion).** On the occupied anisotropic carrier, the source-frozen occurrence holonomy has no nonzero proper invariant subspace of  $\text{STF}_5$ .

This is the weakest invariant-subspace condition that excludes a permanently hidden tensor sector. It is admitted as a 4D-constrained working-completion axiom because the target quantum theory requires the full three-factor carrier; it is not presented as a theorem of the minimal Tau kernel or as an empirical fact.

**Theorem 9** (Terminal rank-five closure in the physical MVP). *Let  $W_{\text{seed}} \subset \text{STF}_5$  be the nonzero tetrahedral rank-three seed image. Under OHC-P1,*

$$\text{span}_p \rho_2(P_p) W_{\text{seed}} = \text{STF}_5. \quad (54)$$

*Consequently MBC-P1/ACT-M1, OHC-P1 and the typed TC-CVP1 restriction select the nondegenerate rooted tensor polarization of Proposition 6. No further continuous tensor selector remains in the physical MVP.*

*Proof.* The orbit span is nonzero and invariant under the occurrence-holonomy group: applying another holonomy element only permutes the subspaces included in its definition. OHC-P1 admits no nonzero proper invariant subspace, so the orbit span must be all of  $\text{STF}_5$ . The rank is therefore five. Proposition 6 then gives a rank-three mixed Hessian with  $c_1 = c_2 = -\mu_0 \neq 0$ , which completes the rooted polarization.  $\square$

The aligned rank-three countermodel remains scientifically useful: it proves that OHC-P1 adds physical content and gives the exact rejection control. The remaining physical question is now whether observations support the OHC-P1 working completion, not whether another formal prerequisite can be introduced.

## 14 Braid holonomy and the body–observer metric handoff

Paper III’s path-holonomy theorem has an exact quantum specialization. Let  $\Sigma_{OS}$  be an independently established oriented two-dimensional effective support, let coincidences be excluded, and let a flat occupied Hermitian channel connection descend to  $\rho_{OS} : B_n \rightarrow U(k)$ . The readout is anyonic if and only if this representation is nontrivial after the source-derived quantizer and terminal null quotient. Abelian statistics has  $k = 1$ ; genuinely non-Abelian statistics requires non-commuting adjacent braid generators. The current Tau source does not force the two-dimensional

support, coincidence gap, flat connection, nontrivial representation or terminal visibility. This conditional theorem must therefore not be read backward as source selection.

The body quotient nevertheless supplies a quantitative, source-side compression question. On the rooted relative carrier write

$$K_{\text{leaf}} = k_R P_R + k_T P_T, \quad k_R = 0.777537432703, \quad k_T = 1.072168783649, \quad (55)$$

where  $P_R$  is the rooted singlet and  $P_T$  the tangential doublet. These are three-dimensional body-stabilization eigenvalues; they do not establish a physical two-dimensional surface. For observer access  $R_O = a_R P_R + a_T P_T$ , exact quadratic compression gives

$$K_{\text{occ},O} = (R_O K_{\text{leaf}}^{-1} R_O^*)^{-1}. \quad (56)$$

The tangential doublet is the softer resolved sector precisely when

$$\frac{a_T}{a_R} > \sqrt{\frac{k_T}{k_R}} = 1.174278004418. \quad (57)$$

Symmetry, rank and positivity do not force this inequality; equal-reduct positive counterparts occur on both sides of the threshold.

The observer first jet can be factored as  $J_O = -H_S L_S$ , with  $H_S = h_R P_R + h_T P_T$  and  $L_S = \ell_R P_R + \ell_T P_T$ . A coefficient-free conditional completion is obtained only after the additional metric-copy postulate

$$H_S = K_{\text{leaf}}, \quad L_S = I. \quad (58)$$

It yields  $K_{\text{occ},O} = K_{\text{leaf}}^{-1}$  and the normalized tangential-to-radial stiffness target

$$\chi_{2D} = \frac{k_R}{k_T} = 0.725200588341, \quad g_N = 1 - \chi_{2D} = 0.274799411659. \quad (59)$$

The present body action and observer functional are a typed direct sum, so their mixed Hessian vanishes. Hence Eq. (58) is not a theorem of the current packet. A minimal stationary graph source would instead have quadratic germ

$$\Phi_J^{(2)} = \frac{1}{2} \langle v_A - U_{BA} c_B, H(v_A - U_{BA} c_B) \rangle, \quad H = U_{BA} K_{\text{leaf}} U_{BA}^*, \quad (60)$$

where the existing carrier maps determine  $U_{BA}$  but not the metric-copy law. We call its seed-level naturality requirement the seed metric-copy functor (SMCF). SMCF may act on every admissible observer–source module without storing concrete future observer values in the universal seed, but one-source identity and the currently proved closure conditions do not entail it. A positive metric-modulus counterfamily proves that independence.

An eligible endpoint would prepare three independently normalized rooted perturbations, measure a fully calibrated static susceptibility matrix  $\chi$ , whiten its nuisance metric, and form  $K_{\text{exp}} = [(\chi + \chi^*)/2]^{-1}$ . SMCF predicts a tangential doublet and the ratio in Eq. (59). A frequency ratio  $\omega_T/\omega_R = 0.851587099680$  follows only if common scalar inertia is independently established. Public braiding, fusion, interference and engineered-spectrum packages audited to date do not provide the required source-owned calibrated  $3 \times 3$  response, so none currently tests SMCF. Arbitrary peak matching is inadmissible.

## 14.1 Cross-observer access is not free terminal relabelling

The joint-terminal analysis of Paper VII-A permits the same parent direction to be quantum-visible for one observer and gravity-visible for another when their source-owned access maps have the required fiber factorization. This is an information-level statement. It does not make “quantum” an arbitrary observer label. The terminal constructed here is typed by its effect and instrument algebra, coherent composition, Born probabilities and no-signalling law.

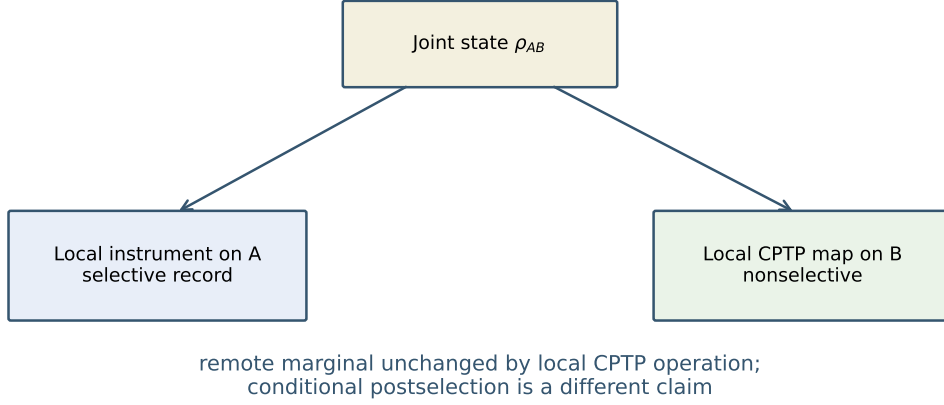


Figure 3: Local deterministic operations preserve the remote marginal. Selective outcomes update conditional states, but the nonselective sum and classical outcome ledger prevent superluminal signalling.

If an injective multiplicative cross-observer map identifies this terminal with another terminal, it must preserve commutators. Hence a genuinely noncommutative quantum fragment cannot be fully identified with a commutative classical-gravity fragment by access change alone. At the more general process level, a full faithful probability-preserving equivalence would transport any classical noncontextual simplex representation back to the quantum fragment. Certified contextuality or incompatibility therefore requires the matched terminal to retain nonclassical operational structure under the same frozen causal, dimension and memory controls.

Sequence dependence alone is insufficient because classical disturbance, backreaction and memory can imitate it. The physical test must reconstruct the common parent fibers, certify operational equivalences independently, reject a bounded classical comparator and transport composition with one source-frozen map. Paper VII-A owns the general multi-terminal theorem; the present paper supplies the quantum signature that any proposed role exchange must preserve.

## 15 Tensor composition and no-signalling

**Theorem 10** (Local CPTP no-signalling). *Assume  $(Q\mathcal{I})$  and a bipartite state  $\rho_{AB}$  on  $\mathcal{H}_A \otimes \mathcal{H}_B$ . For every CPTP map  $\mathcal{E}_B$ ,*

$$\text{Tr}_B[(I_A \otimes \mathcal{E}_B)(\rho_{AB})] = \text{Tr}_B \rho_{AB}. \quad (61)$$

*Thus deterministic local operations on B cannot change the statistics of all effects measured on A.*

*Proof.* Let  $\mathcal{E}_B(X) = \sum_k M_k X M_k^\dagger$  with  $\sum_k M_k^\dagger M_k = I_B$ . For every effect  $E_A$ ,

$$\text{Tr}[(E_A \otimes I_B)(I_A \otimes \mathcal{E}_B)(\rho_{AB})] = \sum_k \text{Tr}[(E_A \otimes M_k^\dagger M_k) \rho_{AB}] \quad (62)$$

$$= \text{Tr}[(E_A \otimes I_B) \rho_{AB}]. \quad (63)$$

Equality for all  $E_A$  proves Eq. (61).  $\square$

A selective instrument outcome is trace-non-increasing. Its normalized conditional remote state may change, but the outcome probability must be communicated classically before it can be used. Summing over outcomes restores the CPTP map and Eq. (61). Figure 3 separates these two statements.

Entanglement is therefore available once the tensor carrier is selected: it is nonseparability of  $\rho_{AB}$ , not hidden classical correlation. What remains open is why the physical parent selects this tensor composition and its occupied entangled states. The no-signalling theorem is conditional on that composition and cannot be used to derive it retrospectively.

### 15.1 Atemporal history refinement and no retro-signalling

Delayed-choice conditioning can be stated without a Parent history store. Let  $(\mathcal{P}, \mu)$  be one source-frozen atemporal Parent dilation and pre-choice weight for the prepared ensemble. For settings  $a, b$ , let

$$X_a : \mathcal{P} \rightarrow \mathbf{X}, \quad D_b : \mathcal{P} \rightarrow \mathbf{D} \quad (64)$$

be stable early and later record maps. These are simultaneous typed maps, not a claim that every incompatible observable has a global noncontextual value. The observer-compatible fibres obey

$$\mathcal{H}_x^{(a)} = X_a^{-1}(x), \quad \mathcal{H}_{x,d}^{(a,b)} = X_a^{-1}(x) \cap D_b^{-1}(d) \subseteq \mathcal{H}_x^{(a)}. \quad (65)$$

This is refinement of a conditional description, not rewriting an earlier Parent state. Stable record invariance separately requires an unconditioned later operation  $\Lambda_b$  to satisfy

$$\Lambda_b^*[f(X_a)] = f(X_a) \quad \text{for every } f \text{ in the early record algebra.} \quad (66)$$

Record invariance alone does not imply no-signalling: a later-setting- dependent preparation weight  $\mu_b$  can change early frequencies while leaving the record map fixed.

With one common pre-choice  $\mu$  and a normalized later-local kernel  $K_b(d|p)$ , the joint law is

$$P(x, d|a, b) = \int_{\mathcal{P}} \mathbf{1}_{X_a(p)=x} K_b(d|p) d\mu(p), \quad \sum_d K_b(d|p) = 1. \quad (67)$$

Therefore

$$\sum_d P(x, d|a, b) = \int \mathbf{1}_{X_a(p)=x} d\mu(p) = P(x|a), \quad (68)$$

independently of  $b$ . In the quantum terminal this is the standard identity

$$\sum_d \text{Tr}[(E_x^a \otimes F_d^b)\rho] = \text{Tr}[(E_x^a \otimes I)\rho]. \quad (69)$$

Conditioning on  $d$  may still expose complementary interference patterns. For example, terms proportional to  $+V \cos \phi_x$  and  $-V \cos \phi_x$  can occur in two postselected subensembles while cancelling in their earlier marginal. Thus later records refine the compatible fibre but do not change the locally accessible earlier statistics or open a future-to-past information channel. Standard quantum mechanics already explains this structure; its compatibility is not evidence for Tau Core.

## 16 Measurement, decoherence and the classical limit

### 16.1 Quantum instrument

A measurement terminal is represented by an instrument  $\{\mathcal{I}_a\}_a$ , where each  $\mathcal{I}_a$  is completely positive and trace-non-increasing, and  $\sum_a \mathcal{I}_a$  is CPTP. Then

$$p_a = \text{Tr} \mathcal{I}_a(\rho), \quad \rho_a = \mathcal{I}_a(\rho)/p_a. \quad (70)$$

This is a consistent terminal record law. It is not yet a proof of a physical parent collapse. The unconditioned state follows the deterministic channel; the conditioned state records the selected outcome.

## 16.2 Internal measurement backaction and the Parent boundary

The instrument formalism describes terminal records, while a separate regular post-body response calculation identifies physical disturbance of an internal target. With the stabilized body held fixed, let  $o$  denote an apparatus or observer coordinate and  $x$  a post-body target coordinate in one supplied  $C^2$  functional. Then

$$D_o x_* = -H_X^{-1} C_{XO}, \quad H_X = D_x^2 \mathcal{A}_{\text{post}}, \quad C_{XO} = D_x D_o \mathcal{A}_{\text{post}}, \quad (71)$$

and every smooth quantum record precursor obeys

$$D_o \Pi_Q^{\text{occ}} = D_o \Pi_Q - D_x \Pi_Q H_X^{-1} C_{XO}. \quad (72)$$

The direct term can change under apparatus context even at  $C_{XO} = 0$ ; only the second term is target-mediated backaction. For a hard outcome map, the exact control is the same-context, clamped-target record, not a comparison with a different apparatus setting. Hence an unchanged outcome cell does not exclude continuous disturbance, and a changed outcome does not by itself identify a nonzero target coupling. Standard quantum mechanics already contains measurement disturbance and instrument conditioning, so this conditional identity is not Tau evidence. It also does not let the observer reselect the frozen body; physical source ownership of  $C_{XO}$ , Nature occupation and a Tau-specific component remain open.

## 16.3 Decoherence is not outcome selection

For a preferred pointer algebra, a dephasing map removes off-diagonal phase witnesses while preserving populations. Markovian semigroups may be written in Lindblad form [3]. Decoherence explains why interference becomes inaccessible to a coarse observer and why stable records can behave classically. It does not select one realized outcome from the ensemble.

## 16.4 Classical regime

The classical limit used here requires all of the following:

- (C1) suppression of phase-sensitive effects on the occupied record algebra;
- (C2) redundant compatible records across accessible observer fragments;
- (C3) a stable approximately commuting pointer algebra;
- (C4) terminal resolutions coarse compared with the action-phase scale;
- (C5) recovery errors below the declared record tolerance.

The limit is therefore observer-relative but not arbitrary: it is constrained by the occupied body relations and by independently testable record stability.

## 17 Three distinct entropies

Information loss in the descent defines a coarse-graining entropy, for example the log measure of a stable fibre or the Shannon entropy of unresolved classes. This quantity is not automatically the von Neumann entropy

$$S_{\text{vN}}(\rho) = -\text{Tr}(\rho \log \rho), \quad (73)$$

and neither is automatically thermodynamic entropy. The identifications require additional claims:

- coarse-graining entropy equals  $S_{\text{vN}}$  only when the quantum state and its spectral probabilities are the sufficient observer record;

- thermodynamic entropy requires a macroscopic state space, conserved loads and an equilibrium or typicality argument;
- entropy production requires irreversible effective dynamics or loss to an inaccessible sector, not merely a many-to-one notation.

Recovery theory makes this distinction operational. Exact sufficient recovery can preserve relative entropy [7]; correctable codes can preserve the logical record even through a noisy physical carrier [6]. Hence information loss in one representation need not be physical destruction of the occupied quantum information.

## 18 Planck action scale

Equation (41) shows what the Planck constant does in this completion. The parent packet supplies one action unit  $U_A$  shared by phase and clock descent. Matching the terminal law to quantum mechanics identifies

$$U_A = \hbar. \quad (74)$$

This gives a structural interpretation:  $\hbar$  is the action resolution that converts source-owned phase transport into the observer's terminal clock and energy units. It is not proved to be a universal smallest parent action, and the numerical SI value is not predicted. A first-principles result would require a source normalization tied independently to the base-seed action and to operational SI standards.

### 18.1 Downstream particle and RG handoff

Technical Paper VII-C owns the scalar-15/type-II source preimage, its class-relative mass/Planck ratio and the exact functional-RG evaluation law. Paper VI does not rederive those results. It uses them only to delimit the field-theory handoff: a selected particle branch may share the source action unit while its occupation, component beta functions and SI calibration remain separate physical obligations. The RG regulator is an analysis scheme, not the observer-source quantizer  $Q_{OS}$  and not a quantum-readout mechanism.

## 19 Quantum field theory recovery and frontier

The finite carrier does not by itself yield QFT, but the theory hub already contains a stronger conditional field packet than this paper previously recorded. Assume that the body descent supplies a globally hyperbolic metric and a Green-hyperbolic free operator in the standard locally covariant framework [10]. Its retarded-minus-advanced propagator has causal support, so spacelike-supported observables commute. On a stationary globally hyperbolic background, assume a positive self-adjoint time-translation generator with no obstructing zero modes and the regularity conditions required by the ground-state construction. The resulting compatible pure quasifree covariance is then Hadamard [11]. Positivity of a generic time-dependent occupied operator alone would not establish this conclusion. The existing graph-normal field/occupation block can then add a smooth coherent excitation without changing the Hadamard singularity class.

For local interaction functionals, renormalized time-ordered products obeying causal factorization generate relative  $S$ -matrices and spacelike-commuting interacting local algebras order by order [12, 13]. Thus the physical MVP conditionally admits the standard architecture of perturbative QFT on curved spacetime, including microcausality and the time-slice interpretation. These QFT results are imported conditional limits, not new Tau derivations. The Tau-specific question is whether one body-derived source packet supplies their metric, state and local-interaction inputs without independent terminal choices. The paired bilocal control shows why locality must remain an explicit source property: the same free reduct admits nonlocal interactions.

This is not a nonperturbative construction and does not determine the interaction density, couplings, renormalization conditions, spin-statistics, gauge/BRST sector or Standard Model spectrum. A generic time-dependent body also need not select a unique vacuum. Quantum stress and its stochastic second cumulant can source a classical body-derived metric in the joint functional, but a path integral over metrics, its gauge complex and a physical inner product are not derived. The current minimum branch is therefore quantum matter/readout on a classical body-derived geometry, with stochastic backreaction allowed; metric quantization remains undecided.

## 20 Standard quantum recovery and Tau-specific discriminator

The recovery result and the novelty claim must be kept separate. Under the frozen physical MVP, the following standard structures are recovered:

- (R1) positive normalized density operators and POVMs;
- (R2) the trace Born rule and coherent interference;
- (R3) self-adjoint unitary transport with action unit  $U_A = \hbar$ ;
- (R4) tensor composition, singlet correlations and the Tsirelson value  $|S_{\text{CHSH}}| = 2\sqrt{2}$  on the selected two-qubit packet [14];
- (R5) invariance of a remote marginal under local nonselective CPTP maps;
- (R6) free algebraic QFT and the perturbative causal construction under the field assumptions above.

These are necessary compatibility results, not Tau-specific deviations.

The proposed Tau discriminator concerns two quantities that would have to be reconstructed by independent routes. It is not yet an executable prediction, because no audited carrier supplies the required independent action-side observable. For one predeclared occupied record pair define

$$x_A := \frac{\mathcal{A}_X^{PC}}{\mathcal{A}_*}, \quad x_Q := d_{U, \text{chord}}^2(\rho_0, \rho_1) = 2 \left( 1 - \sqrt{F(\rho_0, \rho_1)} \right). \quad (75)$$

The common-Gram completion predicts

$$\boxed{x_A = x_Q}. \quad (76)$$

Standard quantum mechanics determines  $x_Q$  from state tomography but has no independent morphological action variable  $x_A$  and therefore does not imply Eq. (76). This cross-domain identity, rather than a modification of the Born rule, is the primary conditional Tau discriminator.

### 20.1 Projective quantum measure versus body volume

The finite common-Gram identity above compares an action separation with an Uhlmann chord on one occupied record pair. It must not be read as a global identification of quantum-state volume with body volume. In the rank-four source/body prototype, CCOC supplies only

$$A_M : V_{OS}^{\text{rel}} \longrightarrow S_B, \quad \text{rank } A_M = 3, \quad (77)$$

while OGAS supplies a compatible signed common line. Their combined rank is four, but no full source-owned map

$$\Xi : T(\mathbb{CP}_s^1 \times \mathbb{CP}_b^1)|_U \longrightarrow E_T \oplus E_S \quad (78)$$

follows. In particular, rank equality does not imply  $\Xi^* d\mu_{h_*} = d\mu_{FS}$ .

If such a connector is independently supplied, preserves the declared splittings and has  $|\det A_M| = g_{FFL}^3$  in the fixed oriented volume frames, then exact measure matching uniquely fixes its common-line Jacobian:

$$\Xi_{\gamma_T} = \gamma_T \oplus A_M, \quad \gamma_T = g_{FFL}^{-3}. \quad (79)$$

This conditional uniqueness removes a possible blockwise gain. It neither constructs  $\Xi$  nor identifies the body volume operationally, and it does not replace the independent finite-action/Uhlmann test Eq. (76). A common-line rescaling leaves the CCOC spatial/contact data invariant while changing four-volume, so physical connector ownership and projective/body measure identification remain upstream open questions. The absolute M4 unit frame is a different issue: Paper V proves that the common clock-gravity-action packet leaves only

$$(L_H, T_H, \mathcal{A}_*) \mapsto (\sigma L_H, \sigma T_H, \sigma^2 \mathcal{A}_*), \quad \sigma > 0, \quad (80)$$

with all dimensionless clock-ruler terminals invariant. One measured anchor chooses an SI representative; this does not construct  $\Xi$ , fix the finite-action/Uhlmann bridge, or determine a dimensionless quantum coupling.

The prediction remains quantitative under certified approximate recovery. If a record map  $\mathcal{C}$  has a recovery  $\mathcal{R}$  satisfying  $\sup_\rho \|\mathcal{R}\mathcal{C}(\rho) - \rho\|_1 \leq \eta$  on the frozen state family, contractivity and the Bures triangle inequality give

$$0 \leq x_A - x_Q^{\text{out}} \leq x_A - [\max\{0, \sqrt{x_A} - 2\sqrt{\eta}\}]^2. \quad (81)$$

Thus recovery uncertainty broadens the equality into a precomputable interval; it is not an adjustable fit.

Writing  $x_A = \epsilon_O^2$ , the common-Gram identity gives the Bures angle  $D_B = 2 \arcsin(\epsilon_O/2)$ . For a constant energy spread under a declared self-adjoint unitary protocol, the standard quantum speed limit then yields

$$\delta t_{\min, O} \geq \frac{2\hbar}{\Delta H} \arcsin \frac{\epsilon_O}{2} \geq \frac{\hbar \epsilon_O}{\Delta H}. \quad (82)$$

The inverse-energy law and Bures speed limit are standard [9, 15]. Tau Core predicts that their distance input is fixed by an independently measured morphological action separation. A timing violation can reject the occupied completion only after Eq. (76), record hold, recovery quality and  $\Delta H$  have been certified independently.

Define the preregistered residual

$$\Delta_{AQ} := x_Q^{\text{out}} - x_A. \quad (83)$$

Exact recovery predicts  $\Delta_{AQ} = 0$ ; approximate recovery predicts the one-sided interval in Eq. (81). A statistically significant value outside that interval falsifies the common-Gram physical MVP for the tested carrier. Passing it supports the shared-source identity but does not uniquely establish Tau Core, because another theory could impose the same relation. The model currently predicts no universal numerical  $\epsilon_O$  or laboratory effect size; those require identification of a physical carrier and independent action reconstruction.

## 21 Public-data audit

We next ask how far Eq. (76) can be tested without a new laboratory protocol. Two public same-carrier datasets already provide nontrivial controls. Tan et al. measured the quantum metric of one superconducting qubit by both sudden quench and weak periodic drive [16]. Extraction from the public vector figures gives 17 matched  $g_{\phi\phi}$  points with Pearson correlation 0.9892, MAE 0.01952 and RMSE 0.02222; nine nearly constant  $g_{\theta\theta}$  points give MAE 0.00959. The observed

cross-protocol offsets are comparable to published decoherence, finite-time and control scales, so they cannot be identified as a Tau gain.

Yu et al. measured metric components and Berry curvature on an NV-centre carrier using distinct parametric modulations [17]. On the seven non-boundary matched coordinates, the pure-qubit identity

$$|F_{\theta\phi}| = 2\sqrt{g_{\theta\theta}g_{\phi\phi} - g_{\theta\phi}^2} \quad (84)$$

has Pearson correlation 0.9933, MAE 0.01545 and RMSE 0.01577 in the public vector-figure extraction. This independently replicates a cross-readout local quantum geometry on a second platform.

These positive results support the standard quantum-geometric recovery and a local same-carrier connector proxy. They do *not* test the finite Tau identity. Every reconstructed axis is a standard quantum-geometric terminal; none is an independently measured once-counted morphological action  $\mathcal{A}_X^{PC}$ . The public figures also do not provide a certified finite Uhlmann endpoint pair with the required recovery ledger. Therefore current public data determine neither  $x_A$  nor  $\Delta_{AQ}$ . This is a finite identifiability no-go, rather than a null Tau result: the datasets validate part of the standard-recovery side but cannot confirm or falsify Eq. (76). The frozen points, provenance manifest and recalculation script are included in the reproducibility package.

### 21.1 Execution-to-process ledger audit

A second bounded census, inherited from Paper V’s temporal laboratory route, asks a distinct question: does a public experimental package join an actually executed, signed and winding-resolved control waveform to physical boundary keys, same-run tomographic shots, a complete QPT/Choi reconstruction and a calibration/nuisance ledger? No package verified on 24 August 2026 passed all of these gates. The public quantum-memory repository is the strongest laboratory near-pass [18]; the NGQG and NMN releases are process-rich but lack the required execution join [19, 20]. QDataSet supplies waveform-rich synthetic evolution, while the IBM pulse archive supplies hardware pulse programs and state-tomography workflows rather than the complete same-run process record [21, 22].

This does not weaken the complete-Choi detection theorem. It prevents a retrospective common-source test: without an immutable join key, permutations of waveform-to-process pairings preserve both released marginals. Combining separate public repositories therefore cannot manufacture the missing incidence. The minimum operational acquisition is one run-keyed execution–boundary–QPT ledger; independent physical ownership of the Tau source port remains an additional upstream condition.

### 21.2 Complete-process/apparatus near-pass and the Nature boundary

A complementary public-data question is now decidable: can one acquire a physical informationally complete micro-process family together with a real apparatus descriptor? Liu et al. answer this weaker question positively [23, 24]. Their checksum-frozen archive contains four IBM device/date/qubit-pair contexts. Each context uses a fixed  $16 \times 16$  two-qubit preparation/effect grid; the accompanying backend snapshot retains chip coordinates, the coupling map, calibration properties and partially public Hamiltonian parameters. Edge, non-edge, idle, remote-gate and parameterized-gate records supply useful standard controls.

This packet does not establish occupation of the Kato branch derived above. Let  $v$  denote a source-frozen morphology/access tangent, let  $j_v$  be its predicted Choi derivative and let  $N$  span the declared standard nuisance tangents. A model-class-relative Nature certificate requires, among its conjunctive conditions, a smooth independently measured constant-rank access-projector family on common run keys, nonzero Kato activation, nuisance novelty  $(I - NN^+)j_v \neq 0$ , held-out mean/covariance agreement, source-mode and internal-support-motion nulls, a source-derived stable quantizer-cell crossing and independent replication. The public backend snapshot is a static

standard hardware description, not such a family. The release also omits shot arrays, repeat-level source–process covariance, source-derived  $Q_{OS}$  cells and a Tau derivative frozen before the terminal process data.

The frozen seven-clause audit therefore yields

|                                      |          |
|--------------------------------------|----------|
| fully passed clauses                 | 0/7      |
| partial physical/method controls     | 3/7      |
| failed or non-evaluable clauses      | 4/7      |
| complete terminal-process occupation | observed |
| Tau Kato micro-kernel occupation     | open     |

The positive acquisition result is consequently method-level: rich physical process data and an independently supplied standard apparatus descriptor can coexist. It is not a Tau detection. In particular, the observed process variation cannot be differentiated backward to define  $v$  or  $j_v$ , and a static coupling graph cannot be silently promoted to the outward projector jet of Eq. (15). This strengthens the empirical handoff while leaving the complete-Choi theorem conditional and Nature occupation open.

### 21.3 Elastic-sheet body/process double reconstruction

A reduced elastic-sheet model provides a non-quantum standard control for the projector-transport logic used above. Two support coordinates  $s = (\theta, \phi)$  define

$$n(s) = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)^\top, \quad P(s) = I - n(s)n(s)^\top, \quad (85)$$

and a positive mass-normalized elastic family

$$K_e(s) = \omega_l^2 P(s) + \omega_h^2 [I - P(s)], \quad H_e(s) = K_e(s)^{1/2} = \omega_l P(s) + \omega_h [I - P(s)], \quad (86)$$

with nonzero gap  $\delta = \omega_h - \omega_l$ . The body route reconstructs the closed-loop Kato transport from the instantaneous spectral projectors,

$$\frac{dU_B}{d\phi} = [\partial_\phi P, P]U_B. \quad (87)$$

For the latitude loop  $\phi : 0 \rightarrow 2\pi$ , its restriction to the low-mode plane is a rotation through the enclosed solid angle  $2\pi(1 - \cos \theta)$ . The occurrence route does not use  $\partial_\phi P$  or the body holonomy. It instead integrates the complete analytic-signal wave process

$$i \frac{dU_{\text{occ}}}{dt} = H_e(s(t))U_{\text{occ}}, \quad (88)$$

removes only the known common phase  $e^{-i\omega_l T}$  after the loop, and takes the polar factor of the restricted map. Standard adiabatic transport then gives

$$\lim_{T\delta \rightarrow \infty} \mathcal{H}_{\text{occ}}(T) = \mathcal{H}_B, \quad \lim_{T\delta \rightarrow \infty} \|(I - P_0)U_{\text{occ}}(T)P_0\| = 0. \quad (89)$$

An executable finite audit fixes  $\cos \theta = 0.75$ , hence body holonomy angle  $\pi/2$ . Its projector-only discretization has matrix error  $1.14 \times 10^{-6}$ . Increasing the loop duration from 10 to 160 in the declared units reduces the independently propagated occurrence mismatch from 0.2498 to 0.01561 and leakage from 0.1206 to 0.00738. At fixed duration, increasing the gap from 1 to 8 reduces the mismatch from 0.1558 to 0.01950. Reverse-loop, zero-area, fast-loop, gauge-covariance and solver-tolerance controls behave as expected. In the exact gapless control  $K_e = \omega_l^2 I$ , so the physical process is independent of  $s$  and its holonomy is identity even though an artificially retained projector path would remain nontrivial.

### 21.3.1 General isolated band and internal dynamics

The preceding scalar-phase removal uses the exact degeneracy in Eq. (86); it is not valid for a generic isolated band. Let  $P(t)$  be any smooth constant-rank spectral projector of  $H(t) = H(t)^\dagger$ , separated from its complement by  $g_{\min} > 0$ . Define the Kato transport and the internal band dynamics by

$$\dot{W}_K = [\dot{P}, P]W_K, \quad i\dot{U}_{\text{dyn}} = P_0 W_K^\dagger H W_K P_0 U_{\text{dyn}}. \quad (90)$$

The standard finite-dimensional adiabatic theorem gives

$$U_{\text{occ}}(T)P_0 = W_K(T)U_{\text{dyn}}(T)P_0 + O(\varepsilon_{\text{ad}}), \quad \varepsilon_{\text{ad}} \sim \sup_t \frac{\|\dot{H}(t)\|}{g_{\min}^2}, \quad (91)$$

up to the usual smoothness-dependent constants. For a fixed path traversed in duration  $T$ , the controlled error is  $O(T^{-1})$ . The context derivative uses calibrated laboratory time; it is not an external time assigned to the atemporal Tau parent. When  $PHP = \omega_l P$ ,  $U_{\text{dyn}} = e^{-i\omega_l T} P_0$ , recovering the earlier scalar formula. Otherwise the physical comparison must independently reconstruct and remove the full matrix  $U_{\text{dyn}}$ , then compare the polar factor of  $P_0 U_{\text{occ}} P_0 U_{\text{dyn}}^\dagger$  with  $P_0 W_K P_0$ .

An exact countercontrol holds the projector path fixed. In an oriented frame  $B = (e_\theta, e_\phi)$  of  $\text{ran } P$ , compare

$$H_0 = \omega_l P + \omega_h(I - P), \quad H_\mu = B \text{diag}(\omega_l + \mu, \omega_l - \mu) B^\dagger + \omega_h(I - P). \quad (92)$$

These families have identical  $P(s)$ , Kato transport and body holonomy but different in-band spectra and complete occurrence propagators for  $\mu \neq 0$ . Thus the body projector does not determine the full process. The executable audit uses  $\omega_l = 1 \text{ s}^{-1}$ ,  $\omega_h = 6 \text{ s}^{-1}$  and  $\mu = 0.17 \text{ s}^{-1}$ . The two spectral projectors agree to  $9.43 \times 10^{-16}$ , whereas their complete band transports at  $T = 160 \text{ s}$  differ by 2.103. For the split band, the full factorization error falls from 0.2656 at 10 s to 0.02210 at 160 s, and the matrix-corrected geometric error falls from 0.2499 to 0.02074. Scalar-only correction retains error 2.102; the exactly degenerate internal propagator recovers the scalar phase to  $6.38 \times 10^{-13}$ .

### 21.3.2 Atemporal body efficacy and terminal loss

The physical-body interpretation does not require a body trajectory in a parent meta-time. Conditional on a source-selected body action, let  $M_*$  be its stationary quotient point and  $K_M = D_M^2 \mathcal{A}_{\text{body}}^{\text{phys}}[M_*] > 0$ . A faithful body descriptor makes this point locally non-gauge inside the frozen source model. A body-conditioned temporal generator still requires a positive comparison form, a typed occurrence representation and a calibrated terminal clock:

$$\hat{K}_M = G_M^{-1/2} K_M G_M^{-1/2}, \quad H_{M,O} = \tau_O^{-1} \rho_\Xi(\Lambda_M, \hat{K}_M) = H_{M,O}^\dagger. \quad (93)$$

The same positive  $\hat{K}_M$  admits both  $\dot{u} = -\tau_O^{-1} \hat{K}_M u$  and  $i\dot{u} = \tau_O^{-1} \hat{K}_M^{1/2} u$ . Thus static body structure does not select one temporal law or absolute rate. Once the self-adjoint exit is independently supplied,  $U_M(t_O) = e^{-it_O H_{M,O}}$  is unitary. For contractive source incidence  $J_S$  and terminal access  $R_O$ , the complete prequantizer transfer  $A_{OS} = R_O U_M J_S$  obeys the exact identity

$$\begin{aligned} I - A_{OS}^\dagger A_{OS} &= I - J_S^\dagger J_S \\ &\quad + J_S^\dagger U_M^\dagger (I - R_O^\dagger R_O) U_M J_S \succeq 0. \end{aligned} \quad (94)$$

Observer attenuation can therefore coexist with lossless full-body transport. Stable cell identification supplies a further resolution loss, not a violation of this identity. For a retained/hidden

split

$$H_{M,O} = \begin{bmatrix} H_v & V \\ V^\dagger & H_h \end{bmatrix},$$

exact block inversion gives

$$P_v(zI - H_{M,O})^{-1}P_v = \left[ zI - H_v - V(zI - H_h)^{-1}V^\dagger \right]^{-1}. \quad (95)$$

For  $z = \omega + i\eta$ ,  $\eta > 0$ , the hidden self-energy satisfies  $-\text{Im} \Sigma_h(z) \succeq 0$ . Hence an effectively damped retained subsystem may simply redistribute amplitude into unobserved body modes. A monotone clock change  $t'_O = f(t_O)$  sends  $H' = (dt_O/dt'_O)H$  and leaves  $U_M$  unchanged, so no parent meta-time is introduced.

The executable four-mode audit gives full unitarity residual  $1.35 \times 10^{-15}$ , source/access loss-identity residuals below  $1.6 \times 10^{-16}$ , Schur-resolvent residual  $1.31 \times 10^{-15}$ , and clock-reparameterization residual  $9.23 \times 10^{-16}$ . One nonzero body variation is invisible to one terminal at  $2.40 \times 10^{-16}$  but changes the stacked transfer by 0.3599. Thus one-observer invisibility does not imply a gauge direction. Operational detection still requires stable-cell crossing, and terminal agreement cannot prove unrestricted body ontology because a direct bypass can reproduce the same records.

### 21.3.3 Common instrument–time source packet

The incidence, access and conservative apparatus completion need not be independent assumptions. After the body has been stabilized and frozen, let one endpoint-blind post-body action supply the positive interface Hessian

$$\mathbb{H}_I = \begin{bmatrix} G_S & -B_{MS}^\dagger & 0 \\ -B_{MS} & G_M & -B_{OM}^\dagger \\ 0 & -B_{OM} & G_O \end{bmatrix} \succeq 0. \quad (96)$$

Its normalized mixed blocks

$$J_S = G_M^{-1/2} B_{MS} G_S^{-1/2}, \quad R_O = G_O^{-1/2} B_{OM} G_M^{-1/2} \quad (97)$$

are contractions by principal Schur positivity. Thus their relative sign, units and magnitude are owned by one source-frozen action two-jet rather than inserted as terminal gains.

A symmetric Hessian does not select phase orientation. If the same ordered packet additionally carries a nondegenerate skew form  $\Omega_M$ , its polar factor

$$S_M = G_M^{-1/2} \Omega_M G_M^{-1/2}, \quad \mathbb{J}_M = S_M (-S_M^2)^{-1/2}$$

obeys  $\mathbb{J}_M^\dagger = -\mathbb{J}_M$  and  $\mathbb{J}_M^2 = -I$ . A source reference stiffness/inertia ratio gives  $\omega_* = \sqrt{\kappa_*/\mu_*}$ , and hence the explicit self-adjoint occurrence generator

$$H_{M,O} = i\omega_* \hat{K}_M^{1/2} \mathbb{J}_M \hat{K}_M^{1/2} = H_{M,O}^\dagger. \quad (98)$$

For any contraction  $C$ , the Julia operator

$$\mathcal{U}_C = \begin{bmatrix} C & (I - CC^\dagger)^{1/2} \\ (I - C^\dagger C)^{1/2} & -C^\dagger \end{bmatrix}$$

is unitary. Applying it to both maps in Eq. (97) and inserting the body propagator gives one conservative complete instrument whose observed corner is exactly  $R_O U_M J_S$ . The stochastic kernel still requires the environment state: if  $y = A_{OS}s + B_E \xi_E$  and  $\mathbb{E} \xi_E \xi_E^\dagger = \Gamma_E$ , then  $N_O =$

$B_E \Gamma_E B_E^\dagger \succeq 0$ . Stable resolution likewise requires a source-owned pointer action: calibrated isolated minima  $p_a$  define the Voronoi partition

$$Q_{OS}(r) = \underset{a}{\operatorname{argmin}} \|C_O r - p_a\|_{K_P}^2$$

away from tie surfaces.

The finite audit passes nineteen checks. The interface-Hessian minimum eigenvalue is 0.4468, complete-network unitarity holds to  $4.59 \times 10^{-16}$ , the measured compression is exact at machine precision and the loss identity holds to  $2.48 \times 10^{-17}$ . Same-reduct countermodels reverse the phase form, rescale the reference inertia, replace the environment state or replace the pointer minima. Therefore the common Hessian derives  $J_S$  and  $R_O$ , but does not by itself select phase orientation, absolute rate, stochastic state, stable quantizer or Nature occupation.

#### 21.3.4 Enriched action–form origin of the instrument packet

The common packet can be moved one layer further upstream. Let the source–body and body–observer defects be

$$\rho_{SM} = L_M m - L_S s, \quad \rho_{MO} = L_O o - T_M m,$$

and let one endpoint-blind post-body action contain their weighted norms plus positive base susceptibilities. Direct expansion gives

$$\begin{aligned} G_S &= K_S^0 + L_S^\dagger W_{SM} L_S, \\ G_M &= K_M^0 + L_M^\dagger W_{SM} L_M + T_M^\dagger W_{MO} T_M, \\ G_O &= K_O^0 + L_O^\dagger W_{MO} L_O, \\ B_{MS} &= L_M^\dagger W_{SM} L_S, \quad B_{OM} = L_O^\dagger W_{MO} T_M. \end{aligned} \quad (99)$$

The resulting Hessian is a positive Gram matrix with the exact form of Eq. (96). Its direct source–observer block vanishes by construction. Hence complete mediation, the mixed-block signs and both contractions in Eq. (97) share one relation-law origin.

A symmetric scalar Hessian still cannot produce phase orientation. Suppose the enriched universal seed owns a seed-odd action potential  $\vartheta_U = \nu_U \vartheta_0$ , with  $\Omega_U = d\vartheta_U$  nondegenerate, and the same source owns a positive energy form  $E_M$ . Then

$$L_U = E_M^{1/2} \Omega_U^{-1} E_M^{1/2}, \quad H_U = i L_U = H_U^\dagger. \quad (100)$$

Energy divided by action has inverse-time units, so this derives the occurrence rate without the separate stiffness/inertia ratio in Eq. (98). Reversing the seed orientation reverses  $H_U$  while leaving the positive interface action unchanged. The orientation and physical energy/action unit bridge therefore remain source inputs rather than consequences of the Hessian.

The environment covariance can be selected uniquely after a source occupation law is supplied:

$$\mathcal{F}_E(\Gamma) = \frac{1}{2} \operatorname{Tr}(K_E \Gamma) - \frac{\Theta_E}{2} \log \det(\Gamma \Gamma_{\text{ref}}^{-1}) \implies \Gamma_E = \Theta_E K_E^{-1}. \quad (101)$$

The functional is strictly convex on the positive covariance cone because its second variation is  $\frac{\Theta_E}{2} \operatorname{Tr}(\Gamma^{-1} \Delta \Gamma^{-1} \Delta) > 0$ . This is simultaneous state selection, not memory or parent-time evolution. Changing  $\Theta_E$  changes the noise kernel without changing the deterministic transfer.

Finally, if distinct prototypes form one finite seed-role orbit  $\{p_a\} = G_P \cdot p_0$ , the single potential

$$\mathcal{V}_P(p) = \lambda_P \prod_a \|p - p_a\|_{G_O}^2 \quad (102)$$

has exactly those orbit points as global minima. Their local Hessians are  $2\lambda_P \prod_{b \neq a} \|p_a - p_b\|_{G_O}^2 G_O > 0$ , so the minima and resulting Voronoi cells are stable. This derives the prototypes from one orbit law, but does not derive which orbit or calibration Nature occupies.

The finite audit passes all 22 checks. The relation-Gram reconstruction residual is  $3.20 \times 10^{-16}$ , the direct source–observer block is exactly zero, generator self-adjointness holds to  $3.15 \times 10^{-16}$ , occurrence unitarity to  $3.73 \times 10^{-16}$ , and environment stationarity to  $2.25 \times 10^{-16}$ . The result conditionally derives the complete T28 instrument from a smaller enriched action–form complex. It does not derive that complex, source exhaustivity, physical calibration or Nature occupation from the minimal Tau base.

### 21.3.5 Operational Nature-class identifiability and representative no-go

The remaining source symbols are not separately identifiable from the terminal law. For every  $c > 0$ ,

$$(E_M, \Omega_U) \mapsto (cE_M, c\Omega_U) \implies i(cE_M)^{1/2}(c\Omega_U)^{-1}(cE_M)^{1/2} = H_U, \quad (103)$$

while simultaneous reversal of  $\Omega_U$  and terminal-clock orientation leaves  $e^{-it_O H_U}$  unchanged. Likewise  $(\Theta_E, K_E) \mapsto (c\Theta_E, cK_E)$  leaves  $\Gamma_E = \Theta_E K_E^{-1}$  unchanged. Environment-frame changes leave  $B_E \Gamma_E B_E^\dagger$  invariant, and the congruence

$$C_O \mapsto TC_O, \quad p_a \mapsto Tp_a, \quad G_O \mapsto T^{-\dagger} G_O T^{-1} \quad (104)$$

preserves every pointer distance and stable cell, up to cell relabelling. Terminal evidence therefore identifies only a terminal-equivalence class, not a literal energy/action, environment or pointer-coordinate representative.

Independent source measurements may split terminally equivalent but physically distinct members. Let  $\mathcal{I}_{\text{joint}}$  stack the complete terminal law with independently reconstructed source records, and let  $\mathcal{G}_{\text{phys}}$  retain only genuine coordinate, frame, label and proved-null gauges after calibration. On a regular frozen candidate family, the exact local criterion is

$$\ker D\mathcal{I}_{\text{joint}}(\mathfrak{s}_*) = T_{\mathfrak{s}_*}(\mathcal{G}_{\text{phys}}\mathfrak{s}_*). \quad (105)$$

The constant-rank theorem then identifies the physical class locally inside that declared family.

The corresponding CIT-NOC7 certificate requires one co-registered physical carrier and complete raw record; an independently perturbed nonzero two-edge relation two-jet; a reconstructed seed-odd action form; independent energy, action and clock standards; at least two controlled environment occupations separating  $K_E$  from  $\Theta_E$ ; independent pointer-orbit and calibration reconstruction; and the quotient-rank, held-out joint-prediction, wrong-family and spectator controls. The finite audit verifies 13 mathematical identities: the terminal witness has rank three with three equivalence directions, while independent source anchors raise the coordinate-fixed rank to six. The current corpus completes none of the seven gates as one co-registered physical packet. Thus the representative no-go and a noncircular declared-class test are proved, but physical class occupation, ambient source exhaustivity and unrestricted Nature uniqueness remain open. A terminal-invisible positive spectator is an exact countermodel to the last implication.

This establishes algorithmic independence and convergence inside one frozen standard model. It does not establish a quantum terminal, a Tau-specific micro-kernel or Nature occupation. A physical plate implementation should therefore freeze the modal projector from independent static spectral maps, drive preregistered slow and fast support/tension loops, reconstruct the full sensor-to-sensor process without importing the Kato generator, reconstruct the matrix-valued in-band dynamics independently unless exact degeneracy is measured, and include reverse, zero-area, gap-closing, path-off and redundant-sensor controls. Only a residual predicted before the run and not absorbed by standard elasticity, damping, actuator response, propagation or sensor calibration could promote the exercise beyond a method control.

## 21.4 Compact audit verdict

A frozen public-carrier census applied five hard gates: same carrier, an  $x_Q$ -capable endpoint distance, an action-like observable, an independently once-counted  $x_A$ , and an uncertainty/recovery

ledger. None of the audited candidates passes the independent- $x_A$  gate. The strongest standard controls confirm ordinary quantum geometry and speed-limit behavior, but they do not supply the cross-domain observable required by Eq. (76). The result is therefore an identifiability no-go, not a negative Tau detection.

Table 2: Compact public-carrier verdict. “Partial” denotes a model-dependent or same-pipeline proxy. The full candidate-by-candidate ledger is preserved in the technical supplement.

| Carrier class                               | same carrier | $x_Q$   | independent $x_A$ | recovery/errors |
|---------------------------------------------|--------------|---------|-------------------|-----------------|
| QGT and tomography platforms                | yes          | yes     | no                | partial         |
| Quantum-speed-limit platforms               | yes          | yes     | no                | yes/partial     |
| Work, calorimetry and source-port platforms | yes          | partial | no                | partial         |
| Optical path-integral data                  | yes          | partial | no                | partial         |

The strongest QOR control, the trapped-atom data of Ness et al. [25], is consistent with the standard speed limit: no one-sided violation among 53 public points exceeds  $1.96\sigma$  (maximum  $1.4465\sigma$ ). The Wen optical reconstruction reproduces the phase control and reveals a standard kernel-morphology amplitude dependence, but the source-frozen Tau curvature candidate is null and the optical action is not an independent Tau body action. These results validate parts of the standard recovery pipeline without evaluating  $x_A = x_Q$ .

The later multimode-fibre method control makes the access distinction exact. In the direct transfer-as-access realization all measured square maps are full rank, hence  $P_O = I_{110}$  and Corollary 1 gives zero Kato transport although the normalized Choi process changes. The current scalar source-connection candidate is likewise projectively null for that non-scalar change. A full independently sourced body connection or a varying, gapped source-owned access subspace would be sufficient in principle; neither is physically realized by the current source packet. Fitting a local generator from the same terminal transfer path would test standard optical factorization, not derive a Tau source arrow.

There is also a structural no-go. Tomography, QGT and fidelity are circular as  $x_A$  estimators; band curvature and state-space QGT are independent in a finite two-band countermodel; energy and fidelity susceptibilities are different spectral moments; and equal local stiffness does not determine a finite Uhlmann chord or loop holonomy. Consequently the minimum admissible reopening packet is

$$\boxed{\{K_A, C_q, \nabla^C, A_*, \text{withheld endpoints, loops and recovery ledger}\}}. \quad (106)$$

The body-side stiffness and its action unit must be frozen before the quantum endpoints are examined.

The internal finite audits sharpen but do not remove this boundary. The inherited CHDF quotient supplies a positive body-side action candidate, and exact recovery selects the identity member inside a bounded depolarizing connector family. Generic CPTP structure does not select that member, and ordinary thermodynamic work is not universally proportional to the Bures chord. Thus the remaining physical question is occupation of one common source packet, not another fit of terminal data.

The complete candidate census, all finite countermodels, the alternative source-port survey, Wen path reconstruction, readiness scores and detailed claim boundaries are retained in the separately compiled *Public-Data and Source-Port Audit for the Tau Core Quantum Terminal*. The scripts and frozen data used by both documents remain in this repository.

## 22 Finite decision protocol

The theory does not need unlimited data collection to test this packet. It needs one source-frozen carrier family on which the following finite gates are independently reconstructed:

- (F1) a stable observer partition and ordered record transitions;
- (F2) a positive response metric  $g$  and phase curvature  $\omega$ ;
- (F3) the lock residual  $r_J = \|(g^{-1}\omega)^2 + \kappa^2 I\|$ ;
- (F4) a phase-sensitive interference witness excluded by the classical same-quotient control;
- (F5) a normalized positive effect ledger testing Eq. (42);
- (F6) a compositional no-signalling test with selective and nonselective controls;
- (F7) a recovery/dephasing control separating transport loss from outcome statistics.
- (F8) independent same-carrier reconstructions of  $x_A$  and  $x_Q$ , with the recovery tolerance  $\eta$  frozen before evaluating Eq. (76);
- (F9) only after that identity passes, an independent  $\Delta H$  and timing test of Eq. (82).

The branch is weakened if  $r_J$  remains nonzero beyond uncertainty, if the phase witness vanishes while the proposed quantum packet is occupied, if probabilities require context-dependent gains, or if a local deterministic operation changes a remote marginal. Passing these tests establishes the finite terminal packet, not its unique parent origin.

## 23 Parent-realization theorem and exact boundary

Let  $\mathfrak{P}_Q$  denote the complete packet consisting of the positive metric, nondegenerate phase form, compatible complex structure, common action unit, separating positive effects, tensor composition and exact recovery on the occupied logical algebra. The present base-seed reduct does not entail  $\mathfrak{P}_Q$ . Keep the stabilized body, occupied P3 algebra, normalized positive states and upstream source data fixed, and consider

$$\mathcal{C}_\gamma(\rho) = \gamma\rho + (1 - \gamma)I/2, \quad 0 \leq \gamma \leq 1. \quad (107)$$

Both  $\gamma = 1$  and  $\gamma = 1/2$  are CPTP descents of the same upstream packet. Orthogonal-state trace distance becomes  $\gamma$ . If a CPTP left inverse existed on the full logical algebra, contractivity of that inverse would require  $1 \leq \gamma$ . Hence  $\gamma = 1$  realizes exact recovery and the bounded A8b connector, whereas  $\gamma = 1/2$  does not. The aligned and irreducible occurrence transports provide an independent equal-reduct pair with tensor-orbit ranks three and five. Therefore

$$B_\tau^{\text{phys}} + s_U \not\Rightarrow \mathfrak{P}_Q \quad (108)$$

under the current axioms.

There is nevertheless an exact sufficient source criterion. Call one occupied once-counted parent representation source-faithful and protected when it is simultaneously: (i) action-isometric with the common Hilbert–Schmidt/Uhlmann unit; (ii) metric-symplectic; (iii) separating on the occupied P3 operator system; (iv) monoidal through two faithful commuting  $M_2(\mathbb{C})$  factors; and (v) exactly recoverable on the designated logical algebra, with loss confined to its sourced realization complement. For such an SFPR $_Q$ , the preceding theorems give respectively the complex structure, GNS/Born pairing, complete effect atlas, tensor locality, exact A8b member and inherited action unit. Thus

$$\text{SFPR}_Q \implies \mathfrak{P}_Q. \quad (109)$$

This is a representation theorem, not an existence theorem. The remaining remaining unrestricted input is physical base-seed occupation of one SFPR $_Q$ ; declaring it would close the MVP conditionally, while deriving it requires a stronger parent action or independent physical evidence.

The recovery component has a sharper physical selector. Let  $\mathcal{C}_O^c$  be a complementary channel of observer descent. If, on an isolated occupied logical algebra,

$$\mathcal{C}_O^c(\rho_L) = \tau_G \quad \text{for every } \rho_L, \quad (110)$$

then the inaccessible realization complement contains no logical-state information. The finite-dimensional information–disturbance/correctability criterion [6] implies a CPTP left inverse of  $\mathcal{C}_O$  on the complete logical algebra. At the same minimal input and output dimension the channel is unitary conjugation. This condition is stronger than stable classical recording: complete dephasing preserves the trace distance of  $|0\rangle, |1\rangle$  but maps the orthogonal phase pair  $|+\rangle, |-\rangle$  to the same state.

A parent-side sufficient realization of Eq. (110) is a source-frozen isolated split  $TM_{\text{occ}} = L \oplus G$  satisfying

$$P_L H_{\text{parent}} P_G = 0, \quad \ell_{\text{parent}} = \ell_L \oplus \ell_G, \quad qquad U_{\text{parent}} = U_L \otimes U_G, \quad qquad \rho_{LG} = \rho_L \otimes \tau_G. \quad (111)$$

Thus morphological isolation means absence of a mixed source incidence able to export logical distinguishability into  $G$ , not absence of the surrounding body. The observed unitary limit of isolated quantum systems requires this law operationally. It is therefore a 4D-constrained isolated-scope selector, not yet a first-principles consequence of the current base–seed action. Turning on the mixed block gives the approximate-recovery regime rather than a contradiction.

### 23.1 Disjoint-source valuation and the isolation theorem

This source-locality proposal should not be confused with standard cluster decomposition or with algebraic-QFT locality. Those are terminal statements about correlations or commuting observable algebras. The valuation below is a stronger parent-side composition axiom for disconnected occupied source incidences; the paper tests its consequences without claiming that standard locality selects it.

The block-zero condition in Eq. (111) follows from one finite parent composition law. Let the occupied source hypergraph split into disconnected components and require the once-counted parent action to be a normalized disjoint-union valuation,

$$\mathcal{A}(\emptyset) = 0, \quad \mathcal{A}(X \sqcup Y) = \mathcal{A}(X) + \mathcal{A}(Y) \quad (112)$$

whenever no occupied primitive source incidence meets both  $X$  and  $Y$ . Induction on the finite connected-component decomposition gives

$$\mathcal{A}(H_{\text{occ}}) = \sum_c \mathcal{A}(H_c). \quad (113)$$

Twice differentiating Eq. (113) yields

$$P_c H_{\text{parent}} P_d = 0 \quad (c \neq d). \quad (114)$$

Equations (112)–(114) therefore prove only the action and Hessian parts of morphological isolation. Exact logical recovery additionally requires the source load, generator and occupied state to factor as stated explicitly in Eq. (111), or an independent proof of the logical non-leakage condition (110). A vanishing mixed Hessian alone does not exclude higher-order, global-constraint or initial-correlation leakage.

The valuation is not implied by once-counting, positivity, source freeze and internal component symmetry alone. A positive parent Hessian may contain

$$\epsilon \langle z_L, \mathbf{1} \rangle \langle z_G, \mathbf{1} \rangle, \quad \epsilon \neq 0, \quad (115)$$

while preserving the two internal permutation symmetries. Equation (115) gives  $P_L H_{\text{parent}} P_G \neq 0$  and violates Eq. (112). The valuation is therefore a real composition law, not a restatement of the weaker source packet.

If a physical hyperedge crosses the  $L|G$  partition, its second variation may generate precisely such a mixed block. Logical information can then enter  $G$ , allowing entanglement, local decoherence and nonexact recovery while the joint parent state remains represented. Exact protection is consequently an isolated-scope theorem, not a universal ban on interaction.

## 23.2 Operational-to-parent two-jet lifting theorem and no-go

The finite valuation defect  $\mathcal{A}(X \sqcup Y) - \mathcal{A}(X) - \mathcal{A}(Y)$  is a scalar and cannot be acted on by a terminal Jacobian. The operationally accessible local object is instead its mixed second variation. For source variations  $u \in E_X$  and  $v \in E_Y$ , define

$$N_{\mathcal{A}}^{X|Y}(u, v) := \left. \frac{\partial^2}{\partial s \partial t} \mathcal{A}(X(su) \sqcup Y(tv)) \right|_{s=t=0} \in E_{\text{cross}} := \text{Bil}(E_X, E_Y; \mathbb{R}). \quad (116)$$

Let  $\mathfrak{D}_{\text{prim}}^{(2)} : E_{\text{cross}} \rightarrow E_{\text{term}}^{(2)}$  be the stacked, source-frozen second-jet descent of all declared primitive physical readouts. This typed map sends a parent mixed bilinear response to the corresponding family of terminal mixed responses.

**Proposition 8** (Joint two-jet faithfulness). *Suppose operationally isolated preparations have vanishing mixed terminal responses,*

$$\mathfrak{D}_{\text{prim}}^{(2)} N_{\mathcal{A}}^{X|Y} = 0, \quad (117)$$

*for every admissible pair, and suppose*

$$\ker \mathfrak{D}_{\text{prim}}^{(2)} \cap E_{\text{cross}} = \{0\}. \quad (118)$$

*Then  $N_{\mathcal{A}}^{X|Y} = 0$ , equivalently  $P_X H_{\text{parent}} P_Y = 0$ , on that occupied scope.*

*Proof.* The operational premise places  $N_{\mathcal{A}}^{X|Y}$  in  $\ker \mathfrak{D}_{\text{prim}}^{(2)}$ , while Eq. (116) places it in  $E_{\text{cross}}$ . Equation (118) makes this intersection trivial.  $\square$

This proposition lifts only the local mixed two-jet. It does not recover the finite valuation law (112): that stronger statement requires all mixed finite-response jets to vanish, or an independent global composition axiom. Nor does the proposition alone imply exact recovery, which still requires the remaining factors in Eq. (111).

The faithfulness premise is necessary for *universal identifiability over the admitted cross-defect class*. Add to a positive valuation Hessian a small positive-preserving cross form  $N$  such that

$$P_L N P_G \neq 0, \quad \mathfrak{D}_{\text{prim}}^{(2)} N = 0. \quad (119)$$

The original and modified parents have identical declared primitive two-jet reducts, while only the first has a zero mixed Hessian. Thus no inference valid for the full admitted defect class can establish parent two-jet isolation while a common cross-defect null direction survives. A narrower physical model may exclude the counterform by an additional source law, so faithfulness is not asserted as a logically necessary axiom of every possible completion.

The present series does not yet establish Eq. (118). A global physical  $\mathfrak{D}_{\text{prim}}^{(2)}$  cannot yet be formed on one frozen domain because full electric/magnetic/radiative gravity, photon and gauge occupation, non-Abelian/chiral matter, interacting multi-system QFT and their common calibration are not all completed. The local six-dimensional Green packet is complete on its own descriptor, but that local rank statement is not global primitive-readout completeness. Thus the exact status is

$$\begin{aligned} &\text{joint two-jet faithfulness} \implies \text{zero parent mixed Hessian}, \\ &\text{zero parent mixed Hessian} + \text{global factorization data} \implies \text{protection}, \end{aligned} \quad (120)$$

with the first premise physically open and an equal-readout countermodel excluding a stronger universal inference from the current reduct. In particular, the absence of a completed global terminal stack establishes non-demonstrability, not a theorem that the physical kernel is nonzero.

The possibility-space intuition requires one further distinction. Let  $C_x^+$  be the forward cone selected by the body order and  $K_x > 0$  the local body Hessian. If the local admissibility law is reversal symmetric, then  $v \mapsto -v$  is an isometry between the forward and backward quadratic acceptance bodies. Orientation and stiffness alone therefore cannot enlarge the future. Let  $R_-$  and  $R_+$  instead denote the source-owned restriction maps imposed on backward and forward compatible extensions. Their fibres are

$$F_x^- = \ker R_-, \quad F_x^+ = \ker R_+. \quad (121)$$

A larger future possibility space means a larger dimension or inherited Hessian volume of  $F_x^+$ , as can occur when seed and current-body data fix backward compatibility more strongly than forward completion.

This asymmetry becomes exact when backward extension must also reproduce the observer's already stable records. Let  $R_{\text{rec}}$  be the corresponding record-agreement map on the current cut and set

$$R_- = (R_+, R_{\text{rec}}). \quad (122)$$

Then

$$F_x^- = \ker R_+ \cap \ker R_{\text{rec}} \subseteq F_x^+, \quad \dim F_x^+ - \dim F_x^- = \text{rank}(R_{\text{rec}}|_{F_x^+}). \quad (123)$$

The second identity is rank-nullity applied to the restricted record map. Consequently the future fibre is strictly larger exactly when a stable record constraint removes at least one otherwise future-compatible direction. At fixed resolution  $\epsilon$ ,  $r$  independent removed directions can increase the operational number of resolved future cells relative to past cells by a factor scaling as  $\epsilon^{-r}$  on a normalized bounded chart. This cell count, rather than a comparison of volumes in different dimensions, is the precise sense in which the future can be much more “fluid.”

No stored parent history is used:  $R_{\text{rec}}$  is a simultaneous stability condition in the observer's current readout quotient. Nor does the observer access actual future outcomes. Equation (123) establishes excess compatible directions, not their quantum coherence; the phase form and future-ROOT bridge remain necessary for that promotion.

Here the *future-ROOT bridge* is the paper's map between an oriented future-compatible body fibre and the logical ROOT quantum carrier. It is not the upstream DA-ROOT/M4TR route-agreement map

$$U_{\text{DA} \leftrightarrow \text{ROOT}} : T_{\text{OS},x}^{\text{vis}} \longrightarrow E_4^{\text{ROOT}},$$

which would identify a derivation-reconstructed regular 4D tangent with the independently constructed ROOT/M4TR Lorentz carrier. Closing either bridge does not close the other. The latter must additionally preserve the physical null quotient, order/time-role line and Lorentz-conformal structure, and its source ownership remains open.

The word “larger” does not mean unbounded. The admissible local future body is

$$\mathcal{E}_x^+(\Lambda) = F_x^+ \cap C_x^+ \cap \mathcal{D}_M \cap \{v : g_+(v, v) \leq \Lambda\}, \quad (124)$$

where  $\mathcal{D}_M$  contains the remaining morphology constraints. If  $F_x^+$  is finite dimensional,  $g_+ > 0$ ,  $\Lambda < \infty$  and the causal and morphology constraints are closed, Eq. (124) is compact and has finite Hessian volume. A finite observer resolution then produces only finitely many stable readout cells. The future is more fluid only relative to the record-constrained past; the morphological body still sets its directions, stiffness, boundary and accessible resolution.

The available concrete body quotient also gives an exact dimensional boundary. The CHDF leaf Hessian is positive on one radial and two tangent directions, hence has real rank three. If that leaf alone were identified with  $F_x^+$ , every linear bridge to the real rank-four ROOT qubit would have rank at most three; an antisymmetric phase form on the leaf has even rank at most

two. Thus the CHDF-leaf-only identification cannot realize the full ROOT carrier. This is not a no-go for the enriched body: the P3/ROOT packet already supplies the conditional rank-four target, but no source-frozen body-first matrix  $R_+$  is currently available on that same domain. The numerical rank of the mixed future-ROOT block is consequently undefined, not zero. The next finite construction is an enriched body/P3 restriction retaining a four-real-dimensional non-gauge, metric-visible sector before the ROOT terminal is applied.

The inclusion  $i_+ : F_x^+ \hookrightarrow T_x M_\tau$  induces

$$g_+ = i_+^* K_x > 0. \quad (125)$$

This is a classical possibility geometry. A symmetric Hessian cannot itself produce phase. If the same source supplies a quotient-basic curvature  $\omega_+$  with

$$(g_+^{-1} \omega_+)^2 = -\kappa^2 I, \quad (126)$$

then  $J_+ = g_+^{-1} \omega_+ / \kappa$  gives the complex carrier and the existing coherent positive-effect theorems apply. The observer accesses a current descriptor of the compatible extension fibre, not an already realized future outcome. Operational future-blindness and no-signalling are therefore preserved. Physical realization of  $R_+$ , its rank advantage and the common source of  $\omega_+$  remain open.

The last open phrase can be sharpened by using the variational boundary structure of one parent action rather than postulating an unrelated phase law. Suppose an ordered differentiable parent action has first variation

$$\delta \mathcal{A}_M = \langle E(M), \delta M \rangle + \int_{\Sigma_O} \theta_M(\delta M), \quad (127)$$

where  $\Sigma_O$  is the current observer cut. On compatible linearized future solutions, the bulk second variation induces the Jacobi metric  $g_+$ , while the antisymmetrized boundary variation induces

$$\omega_+(\delta_1, \delta_2) = \delta_1 \theta_M(\delta_2) - \delta_2 \theta_M(\delta_1). \quad (128)$$

Thus metric and phase have one once-counted variational owner. After removing the gauge/null kernel, assume  $g_+ > 0$  and nondegenerate  $\omega_+$ . With  $A = g_+^{-1} \omega_+$ , polar decomposition gives

$$J_+ = A(-A^2)^{-1/2}, \quad J_+^2 = -I. \quad (129)$$

This conditionally derives the complex carrier without an independent phase gain. It does not imply the stronger scalar lock  $A^2 = -\kappa^2 I$ : that requires all polar mode scales to coincide. The finite certificate with  $g_+ = \text{diag}(1, 1, 2, 2)$  and two equal canonical phase blocks has a valid polar complex structure but mode scales  $(1, 1, 1/2, 1/2)$ . It is therefore an explicit counterexample to deriving one universal action unit from common variational ownership alone. Physical occupation, quotient descent and the equal-mode-scale law remain open.

The equal-mode-scale condition itself has a finite representation-theoretic closure. Let a parent symmetry group  $G$  act irreducibly on the occupied real future fibre and preserve both  $g_+$  and  $\omega_+$ . Then  $A = g_+^{-1} \omega_+$  commutes with  $G$ , and so does the positive self-adjoint polar operator  $|A| = (-A^2)^{1/2}$ . Every eigenspace of  $|A|$  is  $G$ -invariant. Irreducibility therefore forces

$$|A| = \kappa I, \quad A^2 = -\kappa^2 I. \quad (130)$$

This is the required scalar lock, not an additional fitted gain. A real four-dimensional quaternionic certificate has a one-dimensional symmetric commutant and zero lock residual. By contrast, a reducible sum of two phase planes retains scales 1 and 1/2. Thus irreducibility is essential. The current base-seed law has not yet been shown to occupy this irreducible future-fibre orbit, and Eq. (130) fixes only relative mode scales, not the absolute value or SI calibration of  $\kappa$ .

No new irreducible carrier is required for this step. The occupied logical ROOT qubit is a complex two-dimensional, hence real four-dimensional, irreducible  $SU(2)$  module. Let

$$C_{FQ} : F_x^+ \longrightarrow \mathcal{H}_{\text{ROOT}}^{\mathbb{R}} \quad (131)$$

be a source-owned bijective intertwiner preserving the metric and phase forms. Pulling the ROOT action back through  $C_{FQ}$  makes  $F_x^+$  irreducible, so Eq. (130) follows. The finite realified Pauli certificate has bridge rank four, zero intertwining residual and a one-dimensional symmetric commutant.

Equation (131) is not selected by the current reduct. Keeping the stabilized body and complete internal ROOT packet fixed admits both  $C_{FQ} = 0$  and a full-rank equivariant  $C_{FQ}$ . Thus the bridge theorem reduces physical occupation to one typed source incidence but does not derive that incidence from the present base-seed law.

The source of  $C_{FQ}$  can be localized further. Decompose the boundary second variation of Eq. (127) into future and ROOT blocks,

$$\omega_{\partial M} = \begin{pmatrix} \omega_F & \beta_{FQ} \\ -\beta_{FQ}^T & \omega_Q \end{pmatrix}. \quad (132)$$

If  $F_x^+$  and  $\mathcal{H}_{\text{ROOT}}^{\mathbb{R}}$  are equivalent irreducible parent modules and  $\beta_{FQ}$  is a nonzero equivariant intertwiner, then its kernel and image are invariant. Irreducibility forces a zero kernel and full image, so  $\beta_{FQ}$  itself supplies a full-rank  $C_{FQ}$ . Separate bijectivity is not an independent assumption.

The diagonal restrictions  $\omega_F$  and  $\omega_Q$  do not activate this block. A finite antisymmetric family with the same two diagonal phase forms admits both  $\beta_{FQ} = 0$  and  $\beta_{FQ} = \lambda I$ ; for  $0 < |\lambda| < 1$  the latter joint form is nondegenerate and its bridge has rank four. Hence common variational ownership alone still does not force the mixed incidence. This block is distinct from the forbidden logical-environment mixing in Eq. (111):  $\beta_{FQ}$  loads the protected logical carrier from the body possibility fibre, while morphological isolation prevents that logical information from leaking into the inaccessible realization complement.

The previously derived BRAC term does not by itself supply this boundary block. Its morphology-dependent potential is zeroth order, so it has a generically nonzero mixed bulk Hessian on an occupied field but contributes no independent covariant symplectic potential. The required source instead occurs in the covariant kinetic term of the selected joint completion (JCSEL, BRDC and EBRP). On an observer cut its canonical momentum is

$$\pi_{\Phi} = Z_{\Phi} \sqrt{\hbar} n^{\mu} \partial_{\mu} \Phi. \quad (133)$$

For a future body variation  $\delta_F$  and field variation  $\delta\Phi$ , the mixed boundary coefficient contains

$$\beta_{FQ}(\delta_F, \delta\Phi) \supset \delta_F \left( Z_{\Phi} \sqrt{\hbar} n^{\mu} \right) \partial_{\mu} \Phi \delta\Phi. \quad (134)$$

SBTCL595's nonneutral EBRP branch supplies a nonzero occupied positive-frequency field momentum. Hence every non-gauge future mode that changes the densitized normal kinetic metric makes Eq. (134) nonzero. Covariance makes the block equivariant; on the equivalent irreducible future/ROOT sector the preceding Schur argument promotes it to the full bridge.

This proves activation inside the selected covariant working completion. It does not prove that the physical base-seed parent selects JCSEL-BRDC-EBRP, that the realized body is nonneutral, or that every future-fibre direction is metric-visible. The unrestricted physical statement therefore remains conditional, while the internal completion question is closed.

## 24 Descriptor migration is weaker than quantum migration

The smooth common descriptor and its resolved operational record are distinct typed objects:

$$\Xi_{OS}^{\text{cont}} : X_{OS} \longrightarrow Z, \quad D_{OS}^{\text{op}} = Q_{OS}(\Xi_{OS}^{\text{cont}}). \quad (135)$$

For a legacy smooth descriptor  $L : X_{OS} \rightarrow Y$ , an ordinary terminal migration  $L = U \circ \Xi_{OS}^{\text{cont}}$  exists exactly when

$$\Xi_{OS}^{\text{cont}}(x) = \Xi_{OS}^{\text{cont}}(x') \implies L(x) = L(x'), \quad \ker D\Xi_{OS}^{\text{cont}} \subseteq \ker DL \quad (136)$$

on a regular smooth stratum. Equality of global fibre relations is required for descriptor equivalence. Neither equal real rank nor a real-linear isomorphism supplies a quantum equivalence.

Let  $J$  denote the complex structure and let  $\mathfrak{A}$  denote the relevant observable algebra. A quantum migration must in addition satisfy the appropriate typed conditions, for example

$$UJ_{\Xi} = J_L U, \quad U(a^*) = U(a)^*, \quad a \geq 0 \implies U(a) \geq 0, \quad (137)$$

together with normalization. When  $U$  represents an operational quantum channel, complete positivity and trace preservation are required. Exact operational equivalence additionally requires a CPTP recovery on the occupied code, not merely an inverse real-linear map. These conditions preserve the complex phase, effects and ancilla-stable positivity needed by the quantum terminal. They are not implied by Eq. (136) or by the hard-cell partition alone.

For a regular post-body action the active response is fixed by the action two-jet,

$$\Gamma_{\text{resp}} = -H^{-1}B. \quad (138)$$

The SBTCL106 Bregman packet gives  $\Gamma_{\text{resp}} = J_{OS}$  and  $C - B^*H^{-1}B = 0$ . This removes an independent gain but does not select the physical quantum intertwiner. A positive same-reduct counterfamily can retain zero Schur remainder while adding a source-vertical response that does not factor through the declared source. Hence positivity, common mediation and Schur neutrality do not prove complex/\*-algebraic ownership, CPTP realization or Nature occupation.

## 25 What is proved and what remains open

Table 3: Claim ledger. “Conditional theorem” means mathematically derived from the explicitly frozen packet, not established as an unrestricted parent-law realization with physical occupation.

| Claim                                                      | Status and condition                                                                                                                 | Class                                       |
|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| Loss alone implies quantum mechanics                       | False; classical and quantum completions share one finite quotient                                                                   | No-go theorem                               |
| Real-linear descriptor migration gives quantum equivalence | False; complex, *-algebraic, positive/normalized or CPTP-compatible structure is an additional obligation                            | Exact claim boundary                        |
| Regular post-body active response                          | $\Gamma_{\text{resp}} = -H^{-1}B$ , reducing to $J_{OS}$ in SBTCL106; positivity and Schur neutrality do not select the vertical jet | Conditional identity plus same-reduct no-go |
| Complex carrier                                            | Follows from positive $g$ , nondegenerate $\omega$ and $A^2 = -\kappa^2 I$                                                           | Conditional theorem                         |
| Unitary transport                                          | Follows from Paper-V time, one action unit and commuting self-adjoint $K$                                                            | Conditional theorem                         |
| Born and interference                                      | Follows from positive trace effects and coherent composition                                                                         | Conditional theorem                         |
| No-signalling                                              | Follows for local CPTP maps after tensor composition is fixed                                                                        | Conditional theorem                         |

*Continued on next page*

*Claim ledger (continued)*

| Claim                                         | Status and condition                                                                                                                                                                                                          | Class                                                   |
|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| Local resolved transition                     | A nonzero record jet leaves a source-owned stable cell only inside the explicit window of Proposition 1; infinitesimal precursor survival is insufficient                                                                     | Conditional theorem and quantizer nonselection boundary |
| Outward access detectability                  | Every nonzero outward Kato jet changes at least one exhaustive complete Choi record, while probabilities and incomplete records may remain blind; standard-apparatus quotient and physical occupation remain open             | Conditional theorem plus exact controls                 |
| History refinement/no retro-signalling        | Later records define nested Parent preimage fibres; a common pre-choice weight and normalized later-local instrument preserve the early marginal, whereas record invariance alone does not                                    | Conditional theorem and countermodel boundary           |
| Measurement                                   | Quantum instruments are consistent; physical single-outcome selection is not derived                                                                                                                                          | Open physics                                            |
| Classical limit                               | Decohered commuting records give a controlled effective limit; not a fundamental collapse                                                                                                                                     | Conditional limit                                       |
| Planck scale                                  | $U_A = \hbar$ by standard-limit calibration; SI value not derived                                                                                                                                                             | Identification                                          |
| Downstream scalar-15 scale and interacting RG | owned by Technical Paper VII-C; imported only as a field-theory boundary                                                                                                                                                      | not part of the quantum terminal derivation             |
| Current reduct selects the packet             | False: Eq. (107) gives an equal-upstream-reduct counterfamily                                                                                                                                                                 | Exact non-entailment theorem                            |
| Joint source criterion                        | One occupied source-faithful protected representation implies the complete packet by Eq. (109); its physical occupation is open                                                                                               | Conditional representation theorem                      |
| Isolation selector                            | Logical non-leakage in Eq. (110) forces exact recovery; the factorized parent law in Eq. (111) is sufficient but not yet derived from the base-seed action                                                                    | Theorem plus physical-MVP selector                      |
| Disjoint-source valuation                     | Exact finite additivity forces a block-diagonal Hessian, but the Hessian statement alone does not force global dynamics, state factorization or recovery                                                                      | Conditional composition theorem                         |
| Operational parent lift                       | Jointly faithful typed second-jet descent lifts vanishing terminal mixed responses to a zero parent mixed Hessian; it does not derive finite valuation or recovery by itself, and global faithfulness is not yet demonstrable | Conditional local theorem plus identifiability no-go    |
| Future-extension carrier                      | Boundary-fibre asymmetry can enlarge the forward possibility space and the restricted Hessian supplies its metric; phase lock and physical fibre selection remain conditional                                                 | Theorem/no-go plus conditional completion               |
| Record-conditioned fluidity                   | Stable backward record agreement gives $F_x^- \subseteq F_x^+$ with dimension gap equal to the restricted record rank; coherence of the excess directions is not automatic                                                    | Exact rank theorem                                      |
| Bounded future body                           | Positive restricted Hessian, finite action cutoff and closed body/causal constraints make the admissible future compact with finite observer-resolved cell count                                                              | Conditional compactness theorem                         |

*Continued on next page*

*Claim ledger (continued)*

| Claim                                 | Status and condition                                                                                                                                                                                                       | Class                                                |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| Common-action phase source            | Bulk and boundary variations of one parent action conditionally supply $g_+$ and $\omega_+$ ; polar decomposition gives $J_+$ , but equal polar scales and one universal action unit do not follow                         | Conditional theorem plus finite no-go                |
| Irreducible mode-scale lock           | A common parent symmetry preserving $g_+$ and $\omega_+$ forces $ g_+^{-1}\omega_+  = \kappa I$ on an irreducible occupied future fibre; physical orbit occupation and relative mode normalization remain open             | Conditional Schur-rigidity theorem                   |
| Future–ROOT bridge                    | A bijective source-owned metric–phase intertwiner to the logical ROOT qubit transfers its real irreducibility and closes the relative scale lock; zero/full-rank equal-reduct completions prove current nonselection       | Conditional bridge theorem plus exact non-entailment |
| Mixed boundary source                 | A nonzero equivariant future–ROOT boundary block is automatically full rank between the equivalent irreducible carriers; identical diagonal packets admit zero/nonzero blocks, so physical activation is open              | Conditional Schur theorem plus exact counterfamily   |
| Kinetic activation                    | BRAC bulk mixing is the wrong type, but the selected JCSEL–BRDC–EBRP kinetic momentum gives $\delta_F \pi_\Phi \neq 0$ for occupied non-null metric-visible future modes; unrestricted completion realization remains open | Conditional physical-completion theorem              |
| Single-system common source           | Derived inside the enriched MVP plus adopted P3 handoff; unchanged-parent realization remains open                                                                                                                         | Conditional source theorem                           |
| P3 handoff selector                   | Separating records with one common scalar quadratic action force the operator-square and graph-normal law; physical occupation remains open                                                                                | Conditional source theorem                           |
| P3 record rank                        | Primitive and binary families have kernels 56 and 35; grade- $\leq 3$ closure has rank 64; physical ternary occupation remains open                                                                                        | Exact finite rank theorem                            |
| Spectral record atlas                 | Canonical binary effects of all grade- $\leq 3$ words are separating; Frobenius pointer copying alone has rank eight; physical spectral closure follows inside the full S4/S5 MVP                                          | Conditional source theorem/no-go                     |
| Spectral closure selects P3 backwards | False: the involutions used by the closure already assume P3HAND-D1                                                                                                                                                        | Circularity no-go                                    |
| Unique tensor factorization           | Not selected by the simple matrix algebra and normalized trace alone; closed in the physical MVP by MBC/ACT, OHC-P1 and the typed TC-CVP1 polarization                                                                     | Conditional completion theorem                       |
| Tensor selector                       | A regular event incidence plus exact minimal-qubit recovery selects one factor axis; canonical expectation fixes its $M_4$ complement, while recovery error $\eta$ bounds off-factor weight by $\eta$                      | Conditional theorem                                  |
| Clifford-flag realization             | Three oriented orthogonal P3 two-planes plus one line construct the commuting factors by Eq. (47); current source data do not select the flag                                                                              | Conditional source theorem                           |
| Tetrahedral flag candidate            | Opposite edges give the correct three-pair combinatorics, but the current CHDF pair image has rank four and cannot carry the required six independent directions                                                           | Exact inherited rank no-go                           |
| Six-edge completion                   | A positive zero-background [22] fluctuation doublet completes the edge carrier and its common $M_8$ quantum algebra                                                                                                        | Conditional selector                                 |

*Continued on next page*

Claim ledger (continued)

| Claim                                  | Status and condition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Class                                                  |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| Central chirality                      | Both odd $Cl_7$ extensions restrict to the same six-edge $M_8$ terminal; the sign is invisible here and remains open only for a parity-odd/Weyl-sensitive extension                                                                                                                                                                                                                                                                                                                                                                                                           | Exact terminal-blindness theorem                       |
| Physical realization of the completion | Zero/positive [22] stiffness obeys current source data; unrestricted six-edge occupation is not entailed                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Exact independence theorem                             |
| Anti-overfitting status                | The missing module is uniquely [22] and its invariant Hessian has one scalar coefficient; positivity and the common unit remain source-selection tests                                                                                                                                                                                                                                                                                                                                                                                                                        | Structural candidate                                   |
| Occurrence-generated $W_2$             | OHC-P1 forces every nonzero seed orbit to span the full rank-five STF carrier; unrestricted OHC-P1 occupation remains open                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Physical-MVP theorem/empirical boundary                |
| Natural tetrahedral Gram               | Opposite pairs are internally orthogonal but the three pair planes have nonzero cross blocks; the natural split is $3 + (1 + 2)$ , not $2 + 2 + 2$                                                                                                                                                                                                                                                                                                                                                                                                                            | Exact no-go                                            |
| Rooted polarization                    | Both sides restrict to $1 \oplus W_2$ ; a nondegenerate source mixed Gram selects a unique polar isometry, but root covariance alone does not force it                                                                                                                                                                                                                                                                                                                                                                                                                        | Conditional selector/no-go                             |
| Perturbative interacting QFT           | Local causal factorization and Hadamard preparation recover the curved-spacetime architecture conditionally; interaction content and nonperturbative existence remain open                                                                                                                                                                                                                                                                                                                                                                                                    | Conditional standard limit                             |
| Quantum geometry                       | Matter/readout is quantum on a classical body-derived metric with possible stochastic backreaction; metric quantization is undecided                                                                                                                                                                                                                                                                                                                                                                                                                                          | Open richer completion                                 |
| Tau-specific discriminator             | Independent same-carrier reconstructions would obey $x_A = x_Q$ , broadened by Eq. (81) under certified recovery error, only on the occupied bounded sector $0 \leq x_A \leq 2$ ; existence of a matching carrier is proved but its physical realization remains open; exact full-code recovery selects the matching depolarizing member and approximate recovery gives a frozen factor-mixing tolerance; physical occupation of the joint packet remains open; no audited public carrier makes the two axes independently observable, and the QOR timing bound is downstream | Proposed conditional discriminator; not yet executable |

The former physical open arrow is now resolved negatively for the current reduct:

$$B_\tau^{\text{phys}} + s_U \not\Rightarrow \mathfrak{P}_Q. \quad (139)$$

The surviving positive question is narrower: does the physical base-seed body occupy the single  $SFPR_Q$  representation characterized above? Independent post-hoc choices would reproduce quantum notation but not a unified Tau explanation. Physical occupation, rather than another internal connector calculation, is now the primary blocker.

## 26 Occupied-support boundary for the quantum packet

For a positive finite-local quantum candidate the response is

$$e_* = (C - B^\dagger A^{-1} B)^{-1} (k - B^\dagger A^{-1} j).$$

A seed-unloaded split candidate therefore stabilizes at zero; a stable loaded or cross-coupled candidate belongs to generated body support. In particular, the algebraic existence of a positive

([22]) fluctuation sector does not by itself establish occupation. This eliminates inert finite-local extensions, but does not prove that the unrestricted base-seed law supplies the source incidence, metric-symplectic lock, tensor factorization or record structure required by the physical quantum packet.

The operational source-generated-completeness handoff fixes the domain of this statement. Under the enriched one-source premise, the quantum packet is defined on the seed-generated occupied physical quotient; inert ambient summands are true-null presentation surplus. For the complete joint relation/state packet this requires its selector to be constant on complete seed fibres, so that the ideal and faithful state factor through the seed key. The present narrow seed does not prove that factorization. A non-null quantum difference at fixed complete seed key is instead a second source or an incomplete seed descriptor and falsifies the one-source completion.

### Pointed-exterior support inherited from the parent packet

The later parent-packet analysis supplies a sharper support result. On a three-dimensional oriented carrier, one full-rank seed incidence induces the common exterior closure

$$K_{\text{cl}} = \Lambda^2 J_s \oplus \Lambda^3 J_s \quad \text{on} \quad \Lambda^2 E_{\text{ax}} \oplus \Lambda^3 E_{\text{ax}},$$

with ranks three and one and with one Hodge metric and one action unit. If the seed is pointed exterior and the source-fidelity/no-bypass law is occupied, then the induced exterior orbit is faithful and neither degree can disappear. This closes the finite pair/top support needed by the corresponding quantum carrier inside that enriched class.

It does not by itself derive Born probabilities, tensor factorization, no-signalling or interacting QFT: those still require the metric-symplectic, effect and composition assumptions proved conditionally in this paper. Nor does it prove that the narrow physical base-seed reduct owns the pointed exterior grammar. The detailed representation and occupation theorems are owned by Technical Paper VII-B.

### Global-law proof obligation

The quantum packet is conditional on the globally occupied morphological completion. The source-fidelity/Hodge action leaves every positive-Schur extra topology sector unoccupied because its exact reduced source satisfies  $k_{\text{eff}} = 0$ . This closes occupation under that law but not Nature's ownership of the law. Independent component zero-jets are invisible to the local metric-symplectic and Born packet and may reverse global selection. Physical promotion therefore requires a deeper source derivation or a source-frozen cross-terminal observation that distinguishes those laws.

### Observer-relative quantum-gravity access on one complex carrier

The common-carrier interpretation can be stated without identifying quantum and gravity as distinct parent substances. Let  $(V_{\mathbb{R}}, g, J)$  be the realification of the occupied Hermitian readout carrier and let  $e_Q$  and  $t_G$  be, respectively, the metric gradients of one normalized effect probability and one smeared stress functional at the same occupied state. Their normalized mixed response is

$$\sigma_{QG} = \frac{|g(e_Q, t_G)|}{\|e_Q\|_g \|t_G\|_g}.$$

It lies strictly between zero and one exactly when the two gradients are neither orthogonal nor collinear. A stable coarse graining contracts the effect span, so the component of a fixed stress direction visible through quantum effects cannot increase. The same parent direction may therefore be predominantly quantum-visible for one observer and predominantly gravity-visible for a coarser observer without conversion between terminal ontologies or creation of new information.

The complex structure supplies equal-metric quadratures but does not select a numerical overlap:  $t_\theta = \cos \theta e_Q + \sin \theta J e_Q$  preserves  $(g, J)$  and realizes every value  $|\cos \theta|$ . Hence neither

$1/\sqrt{2}$  nor any previously derived dimensionless stiffness ratio is a quantum-gravity prediction. Physical promotion requires source-owned effect and stress incidences on one occupied carrier. Standard QFT can also realize this tangent geometry; the result is a conditional information architecture, not a Tau-specific signal.

## 27 Mixed-Hessian versus joint quantum-gravity excess

### 27.1 Inherited common-source transport and action scale

On a faithful occupied source quotient, let  $W_Q$  and  $W_G$  realize the same positive source generator  $K_*$  in the quantum and gravity images. Then

$$T_{G \leftarrow Q} = W_G W_Q^{-1}, \quad T_{G \leftarrow Q} K_Q = K_G T_{G \leftarrow Q} \quad (140)$$

is the unique transfer compatible with both source realizations. If  $W_Q^\dagger h_Q W_Q = W_G^\dagger h_G W_G = K_*$ , it is also an isometry:  $T_{G \leftarrow Q}^\dagger h_G T_{G \leftarrow Q} = h_Q$ .

The quantum phase normalization in the declared common-action completion fixes that shared unit as

$$\mathcal{A}_* = \frac{S_0}{2\pi} = U_A = \hbar. \quad (141)$$

This strengthens the interpretation of  $U_A$ : it is not an independently attached quantum constant once the common completion and standard phase limit are fixed. It remains a conditional calibration. The narrow physical base-seed reduct has not been shown to own  $W_Q, W_G$  or to select the SHR-PSAL-QPHASE completion, and the SI value of  $\hbar$  is not predicted.

Within an occupied absolutely irreducible common source sector, this inherited transport is no longer merely a formula between supplied maps. Multiplicity-one source equivariance makes each nonzero quantum or gravity map scalar-unique; cyclic no-bypass occupation excludes the zero map, the common pullback metric fixes the magnitude, and a source-owned oriented anchor fixes the remaining sign. The resulting transfer is therefore the unique source-natural isometry inside that enriched packet. Zero occupation, repeated equivalent copies and an unfixed anchor are explicit controls. Physical base-seed occupation of the packet remains open.

The common post-body jet itself may be represented by one endpoint-blind Schur-neutral relation-defect square without changing the stabilized seed-body minimum. This removes a further scalar-action ansatz freedom, but does not derive the quantum label: rotated terminal projectors preserve the same scalar action and change the named terminal responses. A complete graded source characteristic, including its upstream role representation, is therefore the minimal sufficient object in the declared class.

The pointed exterior/CAR source compresses this condition further. Its empty root is cyclic on the onto no-bypass orbit, its full matrix representation is absolutely irreducible, and no spectator copy remains. A source-owned oriented two-role plane would therefore supply one quantum and one gravity copy, after which the common product metric closes the normalized Schur fanout. Without that typed plane,  $U(2)$  role rotations preserve the untyped source action and metric while changing the named terminal projectors; zero legs and repeated copies remain allowed. The remaining upstream question is thus ownership and nonzero occupation of one oriented Q/G role incidence, not a new quantum gain or action constant. In the declared enriched P3 plus Tau-local composition class, the ordered nonzero cotangent ports already supply that oriented role plane and local tensor factorization supplies one source copy per role. The hidden-diagonal control shows that full Q/G marginals without this factorization still describe only one constrained copy. Hence the unrestricted blocker is joint physical realization of the known P3 port packet and locality law, not an additional quantum connector. The pre-existing once-counted component action  $\hat{\Gamma}_{\text{CLP3}} = \sum_C \hat{\Gamma}_C$  realizes those two premises together at every attainable global zero. Hence the pointed-CAR, port-factorization and Schur results close the conditional Q/G transport, and the faithful phase branch fixes  $\mathcal{A}_* = \hbar$ , without an independent quantum gain. Equal-reduct hidden-global and

missing-port controls prevent promoting this completion theorem to physical parent selection. Nor can the physical question select one preferred off-shell representative:  $\Gamma_{\text{CLP3}} + \lambda \Gamma_{\text{CLP3}}^2$  preserves the occupied minimum, physical Hessian and typed fanout packet. The identifiable target is the resulting Q/G completion class. Hidden-global and missing-port controls alter its defining data and therefore remain genuine class falsifiers.

The full-4D scoring compiler adds a guardrail to the common-carrier programme. A nonzero parent mixed Hessian, including a mixed effect–stress response, is not by itself a departure from standard quantum theory or standard gravity. For the observer-metric branch the source-frozen comparison is

$$E_K = (K_{HH} - K_{\text{std}}) - CK_{VV}^{-1}C^\dagger,$$

and a graph-normal auxiliary completion realizes  $C \neq 0$  with  $E_K = 0$ . Any future quantum–gravity discriminator must therefore push the complete nonzero excess through independently fixed quantum and gravity terminal maps, retain their full cross-covariance, and use a held-out fixed-hypothesis score. The present quantum packet, QOR construction and public-data audits do not supply such a physically occupied full-4D packet and are not rescored here.

## 28 Product-free P3 falsification boundary

For two independently reconstructed occupied P3 roles, the inherited source-whitened law gives

$$C_a^2 = C_b^2 = I, \quad \{C_a, C_b\} = 0.$$

Hence every real unit vector  $n = (n_a, n_b)$  obeys  $(n_a C_a + n_b C_b)^2 = I$ . Positivity of a normalized state then yields  $|n_a \langle C_a \rangle + n_b \langle C_b \rangle| \leq 1$ ; maximizing over  $n$  proves the product-free necessary condition

$$\boxed{\langle C_a \rangle^2 + \langle C_b \rangle^2 \leq 1.}$$

A resolved violation on source-frozen independently typed roles rejects the minimal P3 completion. Satisfaction does not confirm Tau, because an experimenter-chosen Pauli pair satisfies the same identity in standard quantum mechanics. The role maps must therefore be reconstructed before and independently of the target relation. The updated Technical Paper VII-B audit finds zero eligible packets among 31 public candidates; this is an identifiability/acquisition result, not a quantum anomaly or a Tau pass.

### Actional-closure handoff to the quantum terminal

For a physically owned ordered Parent-action packet, ACT-PCLOSE1 supplies the common regular response  $D_S Z_* = -H_Z^{-1} B_{ZS}$  and hence the quantum output as the fixed descendant

$$O_Q = Q_{OS,Q} \circ T_Q \circ P_Q \circ \Xi(Z_*).$$

This removes an independent quantum response fit, but it does not derive the metric–symplectic lock, effects, Born normalization, tensor composition or no-signalling assumptions of the quantum packet. Kernel completeness of the full terminal stack is only a local reconstructibility statement; by itself it neither produces quantum coherence nor proves Nature occupation.

## 29 Conditional GNS/Gram quantum representation

Once the observer–source descent has independently supplied a star algebra  $\mathcal{A}_{OS}$ , a positive state  $\omega_{OS}$  and a probe family  $x_a$ , the GNS construction gives normalized rays and the correlation matrix

$$G_{ab}^{OS} = \frac{\omega_{OS}(x_a^* x_b)}{\sqrt{\omega_{OS}(x_a^* x_a) \omega_{OS}(x_b^* x_b)}}. \quad (142)$$

Hermiticity, positive semidefiniteness, local ray-phase freedom and Bargmann loop phases then follow exactly. This is a clean representation theorem and is compatible with the standard quantum leaf already contained in Tau Core. It is not, by itself, a Tau-specific departure from quantum mechanics.

The representation must not be confused with the operational quantum derivation. A noninjective or finite-resolution readout can place distinct Parent configurations in one ordinary outcome class, but this alone does not give coherent superposition, a complex linear effect algebra, Born probabilities, tensor composition or no-signalling. Those require the source-owned metric-symplectic/effect/tensor packet and stable quantizer premises developed in this paper. Likewise a Bargmann phase is a projective holonomy only on a supported loop; it is not automatically an anyon braid, a gravitational phase or a new observable.

The precise implication is therefore

$$\begin{aligned} (\mathcal{A}_{OS}, \omega_{OS}, \{x_a\}) &\implies (G^{OS}, \text{projective phases}), \\ (G^{OS}, \text{finite loss}) &\not\implies (\text{Born rule, tensor locality, no-signalling}). \end{aligned} \quad (143)$$

Physical promotion still requires source ownership, the quantum terminal map and Nature occupation; endpoint agreement cannot supply them backward.

### 30 Common-spectrum origin of the expansion-conditioned Q filter

The ECTR specialization gives a non-post-hoc origin for one positive Q-leg filter. After the spatial body symbol has been frozen, assume the source-normalized Legendre–Riesz dual lock

$$G = \frac{K_{\text{src}}}{2A_*} = k_{c,O}^2 A_{0,M}^{-1} > 0. \quad (144)$$

The observer/context crossover  $k_{c,O}$  is source-derived and frozen before terminal comparison. The Q leg becomes

$$\boxed{F_{Q,O}(a) = (I + a^2 G)^{-1/2}}. \quad (145)$$

For positive self-adjoint  $G$ , functional calculus gives the exact subordination identity

$$F_{Q,O}(a) = \frac{1}{\sqrt{\pi}} \int_0^\infty t^{-1/2} e^{-t} e^{-a^2 t G} dt = \int_0^\infty e^{-\sigma G} d\mu_a(\sigma), \quad (146)$$

where  $d\mu_a(\sigma) = e^{-\sigma/a^2} d\sigma / (a\sqrt{\pi\sigma})$  is normalized and has mean  $a^2/2$ . Consequently the ECTR resolvent filter and the CSSR heat family can be two functional-calculus presentations of one source operator rather than curves matched after observing a terminal endpoint.

This common spectrum does not identify terminal semantics. The coframe is a metric descendant, whereas Eq. (145) enters a Q-typed terminal only after the independent complex phase, effect algebra, composition/instrument and source-derived stable quantizer of this paper are supplied. The continuous filter is not itself the operational quantization mechanism. Nor does attenuation on a fixed geometry imply expansion. A metric statement requires the separately typed common coframe solder and a positive occupied access rate.

Thus expansion is a source-soldered argument of the already selected Q filter, not the origin of quantum mechanics. The metric-scale and terminal-weight readouts may share one upstream source, while observed  $a_O$  alone cannot select  $G$ ,  $k_{c,O}$ , the quantizer or the Q/G operational signatures.

Finally, the dual lock is conditional. Holding the same seed, body and coframe while changing the source coevaluation replaces  $G$  by  $C^*GC$  in the corresponding whitened coordinates. Thus the present Tau source has not yet selected the Q filter of Eq. (145), its nonzero access or its Nature occupation. Paper VI inherits an exact Q-terminal specialization, not a proof of universal expansion or a Tau-specific quantum anomaly.

## 31 Observer realization: inherited results and physical boundary

The current observer programme distinguishes a conditional mathematical construction from an occupied physical measuring system. The source order remains base plus one universal seed, stabilized morphological response, observer–source restriction, shared descriptor, and resolved terminal readouts. Neither a detector nor an empirical residual constructs the parent body backwards. Dynamics below concern downstream subsystems only.

The canonical observer-origin derivation, Sections 22–26, owns the following results. The existing joint source action fixes static and frequency-dependent response together. Variation of the BRAC potential couples to the field square; a linear field coupling is a distinct comparator unless derived by expansion about an occupied background. Fixed canonical means select a unique coherent state in a finite positive harmonic packet only if there is no excess fluctuation energy. This is a free/asymptotic preparation statement, not the ground state of an interacting detector. For the existing coupled harmonic pair, the joint ground is pure but either local subsystem is mixed, with its covariance fixed by the complete joint Hamiltonian.

These are conditional derivations and internal numerical checks. The actual observer–source context, physical preparation process, kinetic/energy calibration and stable operational resolution remain unproved. The existing ORA-T1 selector conditionally chooses stable quantizers when the occupied context, admissible partition family and positive closure barrier are supplied; it does not establish those physical premises. A computed ground state is not a preparation mechanism, and a covariance or optimal abstract measurement is not an implemented stable quantizer  $Q_{OS}$ .

**Ownership and use.** The central observer-origin note owns the new contact, state and covariance derivations; the body-readout technical companion carries the consolidated mathematics. Foundation Papers IV, V and VI retain physical-selection, clock-descent and quantum-readout ownership, respectively. This paper imports only the consequences within its scope. No observational dataset, baseline, frozen coefficient, score or empirical conclusion is revised by this update. The reproducibility supplement records the source hashes and the exact conditional/physical distinction.

**Consequence for this paper.** The occupied observer–source context is relational instance data, not a uniquely selected observer identity inferred from the universal seed alone. Physical source realization and record preparation remain separate from conditional closure.

### 31.1 Inherited contact and the occupied-state distinction

For a positive symmetric three-dimensional quotient  $K$ , define  $t_K = \text{tr } K$  and  $s_K = \text{tr } K^2$ . The BRAC shape invariant and its differential are

$$I_K = 9s_K/t_K^2 - 3, \quad (147)$$

$$dI_K[\delta K] = \text{tr} \left[ (18K/t_K^2 - 18s_K \mathbf{1}/t_K^3) \delta K \right]. \quad (148)$$

If  $K = (LH^{-1}L^*)^{-1}$  on the nonsingular leaf, then  $\delta K = -K\delta(LH^{-1}L^*)K$ . Common scaling and orthogonal frame rotation have zero  $dI_K$ ; the existing spectral-shape control has  $I'_K(0) = -1.06341377025\dots$ . This does not identify its coordinate with a physical apparatus displacement.

For one real canonically normalized field at fixed coframe and length,

$$S_K = -\frac{1}{2} \int \sqrt{-g} \frac{I_K}{\ell_H^2} \phi^2 d^4x, \quad b_j = \ell_H^{-2} dI_K[\partial_{a_j} K]. \quad (149)$$

The corresponding potential interaction is  $H_{\text{int}} = \sum_j a_j B_j$  with  $B_j = \frac{1}{2} \int b_j \phi^2 d^3x$ . The actual subsystem pullback, kinetic terms and calibration remain required. For a free real scalar in a stationary inertial vacuum the Wick-square correlation is  $2W^2$ , so this contact has a two-quantum threshold  $2\nu_B$ , not the linear comparator's  $\nu_B$ . With  $c = \hbar = \ell_H = 1$ ,  $m_0 = 0$  and

the source CHDF value  $I_K = 0.0610078751932586$ , the thresholds are 0.24699772305278 and 0.49399544610556. The previous comparison gaps 0.35 and 0.15 therefore have different support in the two contact models. Neither threshold is asserted for arbitrary trajectories or occupied states.

For a supplied coherent background, mean force and connected covariance follow from the same contact:

$$\langle B_i \rangle = \frac{1}{2} \int b_i \phi_{\text{cl}}^2 d^3 x, \quad (150)$$

$$C_{ij}(t, t') = \int d^3 x d^3 y b_i(x) b_j(y) \left[ \phi_{\text{cl}}(t, x) \phi_{\text{cl}}(t', y) W(t, x; t', y) + \frac{1}{2} W(t, x; t', y)^2 \right]. \quad (151)$$

Vacuum Wick ordering here is a comparison prescription, not a derivation of the physical stress renormalization. A nonzero classical mean does not by itself select this quantum state. The kernel is a free interaction-picture input, generally nonstationary, not an all-orders interacting bath law.

### 31.2 Conditional state selection and its resource requirement

For a supplied finite positive oscillator packet, fixed means  $\text{Tr}(\rho c_r) = \alpha_r$  imply

$$\langle H_{\text{exc}} \rangle = \sum_r \hbar \omega_r |\alpha_r|^2 + \sum_r \hbar \omega_r \langle (c_r - \alpha_r)^\dagger (c_r - \alpha_r) \rangle. \quad (152)$$

The second sum is nonnegative. Equality with the classical mean energy forces the state into the common kernel of all  $c_r - \alpha_r$ , hence the unique coherent state. With excess energy  $\Delta E$ ,

$$1 - \langle \alpha | \rho | \alpha \rangle \leq \frac{\Delta E}{\hbar \omega_{\min}}. \quad (153)$$

Displaced number states have the same means and higher energy, providing an explicit counterexample when the zero-excess condition is omitted. Ground-state minimization does not provide autonomous preparation. For energy-calibrated amplitude  $\psi_E$ , the source-preserving displacement is  $\alpha = (2\hbar\Omega)^{-1/2} \psi_E$ . If the input is action-normalized, the energy/clock conversion remains necessary; it cannot be replaced by setting every mode frequency to one.

For a declared common-frame tetrahedral coherent code, equal priors and  $\alpha_i = \sqrt{N} n_i$ ,  $n_i \cdot n_j = -1/3$ , the off-diagonal overlap is  $s = \exp(-4N/3)$ . The exact optimal single-shot discrimination probability is

$$P_{\text{opt}} = \frac{[\sqrt{1+3s} + 3\sqrt{1-s}]^2}{16}. \quad (154)$$

The square-root measurement attains this value. With  $S = \sqrt{G}$  and  $d = S_{ii}$ , the dual operator  $Y = dS/4$  has matching trace and  $Y - v_i v_i^*/4 \succeq 0$  for columns  $v_i$  of  $S$ , proving optimality. Uniform pure loss replaces  $N$  by  $\eta N$  only in that declared channel model. This coherent code is not the earlier one-particle leaf encoding; the bound does not construct a physical receiver or stable  $Q_{OS}$ .

### 31.3 Joint coupling fixes local uncertainty

The existing two-copy action, expanded at an already aligned local well, has  $H = \begin{pmatrix} D & -C \\ -C^T & D \end{pmatrix}$ ,  $D = e_*(S_i + 16\mathbf{1})$ ,  $C = 16e_*\mathbf{1}$  and  $S_i = 3P_i + 4(\mathbf{1} - P_i)$ . Given a positive kinetic matrix  $G$ , canonical quantization and joint harmonic ground preparation, put  $W = (G^{-1/2} H G^{-1/2})^{1/2}$ . Normal-mode diagonalization gives

$$V_z = \frac{\hbar}{2} G^{-1/2} W^{-1} G^{-1/2}, \quad V_p = \frac{\hbar}{2} G^{1/2} W G^{1/2}. \quad (155)$$

For equal inertia  $I_0$ , a ray of curvature  $s_0 = 3, 4$  has frequencies  $\omega_c^2 = e_* s_0 / I_0$  and  $\omega_r^2 = e_*(s_0 + 32) / I_0$ . Its local symplectic eigenvalue is

$$\nu_{s_0} = \frac{1}{4} \sqrt{2 + \omega_r / \omega_c + \omega_c / \omega_r} > \frac{1}{2}. \quad (156)$$

The joint state is pure but the equal-inertia three-mode local purity is 0.627817314635553. The local position covariance also equals

$$V_x = \frac{\hbar}{\pi} \int_0^\infty [D + I_0 \xi^2 \mathbf{1} - C(D + I_0 \xi^2 \mathbf{1})^{-1} C^T]^{-1} d\xi. \quad (157)$$

Thus the full frequency-dependent Schur kernel, not its static value followed by a local vacuum assignment, determines the fluctuations. For dimensionless coordinates,  $H$  has energy units,  $G$  energy-times-time-squared units, and  $\hbar / \sqrt{I_0 e_*}$  controls fluctuation size.

The unit-scale Gaussian has only 0.18974552836899916 probability inside the previously declared radius  $\sqrt{8/3}/4$  about the chosen vertex. That control does not certify localized records. The state is a local quadratic-well ground, not the full nonlinear multiwell ground or the outcome of a preparation process. Its copies have not been physically identified with the BRAC observer. An adopted acceptance ball is not a source-selected stable quantizer or a measured efficiency.

**Scientific status.** The canonical calculations supply exact conditional identities and finite numerical controls, not Nature occupation. Physical observer identification, preparation, nonlinear record stability and calibrated operational resolution remain open. Standard quadratic-detector, coherent-state and coupled-Gaussian physics supply comparison methods, not evidence of a Tau-specific effect.

## 32 Finite source-to-readout implementation and its scope

The current finite laboratory is a reproducible implementation, not a new Nature-selection theorem. The complex finite carrier and its phase orbit do not alone select CCR, Fock structure, a Born instrument or physical records. The lab's Maxwell/Fock and absorption routes are explicit standard-quantum completions; the historical joint preparation/readout calibration is a classical programmed adapter, not independent seed validation.

**Ownership and reproducibility.** The seed-body companion owns the finite minimum/Hessian example, the body-readout companion its ordered access and contact reduction, and the terminal companion its conditional propagation boundary. The physical-source papers retain the source-selection obligation. Code, declared priors, adverse controls and provenance are available at [https://github.com/tau-core-research/tau-core-theory/tree/main/experiments/seed\\_evolution\\_lab](https://github.com/tau-core-research/tau-core-theory/tree/main/experiments/seed_evolution_lab). Numerical tests are internal evidence only; no published empirical score is changed.

## 33 Conclusion

Observer-relative information loss has a precise but limited role. It creates stable operational classes and can make microscopic discreteness coexist with macroscopic continuity. It does not create quantum mechanics on its own. A genuine quantum terminal requires a phase-capable metric-symplectic source packet, positive effects and coherent composition. Once these are frozen, the complex carrier, unitary transport, Born/interference law and local CPTP no-signalling follow with no terminal fit. OHC-P1 closes the continuous tensor-polarization selector inside the physical MVP, and the inherited Hadamard/causal-factorization packet extends the standard recovery to perturbative curved-spacetime QFT under explicit field assumptions. Within the resulting tensor

packet, exact minimal-qubit recovery, regular observer-event incidence and canonical conditional expectation jointly fix the contextual logical factor and its realization complement. Approximate recovery supplies a quantitative stability bound. The unselected central Clifford sign does not affect this even quantum terminal.

The local interface is now sharper. A differentiable oriented precursor can survive geometrically while remaining unresolved inside the observer's stable zero cell. Complete record nonannihilation and the source-owned cell together fix the first sufficient exit scale. A nonzero outward access-projector jet cannot be hidden from an exhaustive complete Choi instrument, although its instantaneous probabilities can remain unchanged. Separately, later records intersect rather than overwrite observer-compatible Parent fibres; common preparation and normalized future-local instruments preserve the earlier marginal. These results close consistency interfaces, not physical packet occupation or a Tau-specific delayed-choice signal.

The paper's distinguishing proposal is not a modified Born rule or an unexplained timing delay. It is the common-Gram identity  $x_A = x_Q$ , with a recovery-error envelope and a downstream QOR speed limit. The public-data audit shows that this proposal is not yet executable because no carrier makes  $x_A$  and  $x_Q$  independently observable. Paper VI is therefore a conditional quantum foundation with an operationally specified future falsification target, not a completed quantum theory of the physical universe. Unrestricted physical occupation, carrier identification, effect size, nonperturbative QFT, metric quantization and the Standard Model remain open.

The scalar-15 and RG packet owned by Technical Paper VII-C sharpens the field-theory hand-off without changing this verdict. It does not prove that the unrestricted base-seed parent occupies that branch, and it does not turn an RG cutoff into the mechanism of observer-relative quantum readout.

The isolation result is correspondingly conditional but sharper than a bare block-zero assumption. Disjoint-source valuation forces action additivity and a zero mixed Hessian, while a crossing hyperedge gives the local interaction boundary. A typed, jointly faithful second-jet descent can lift vanishing operational mixed responses back only to that parent two-jet statement. Exact recovery still requires factorized loads, generators and occupied states, and finite valuation remains a separate global source law. Since the known terminal families do not yet form one complete physical stacked map, even the local unrestricted parent lift remains unproved rather than silently assumed.

## References

- [1] W. Forrest Stinespring. Positive functions on  $c^*$ -algebras. *Proceedings of the American Mathematical Society*, 6:211–216, 1955.
- [2] Karl Kraus. General state changes in quantum theory. *Annals of Physics*, 64:311–335, 1971.
- [3] Goran Lindblad. On the generators of quantum dynamical semigroups. *Communications in Mathematical Physics*, 48:119–130, 1976.
- [4] Armin Uhlmann. The transition probability in the state space of a  $*$ -algebra. *Reports on Mathematical Physics*, 9:273–279, 1976.
- [5] Andrew M. Gleason. Measures on the closed subspaces of a hilbert space. *Journal of Mathematics and Mechanics*, 6:885–893, 1957.
- [6] Emanuel Knill and Raymond Laflamme. Theory of quantum error-correcting codes. *Physical Review A*, 55:900–911, 1997.
- [7] Denes Petz. Sufficient subalgebras and the relative entropy of states of a von neumann algebra. *Communications in Mathematical Physics*, 105:123–131, 1986.

- [8] Michael Kolodrubetz, Dries Sels, Pankaj Mehta, and Anatoli Polkovnikov. Geometry and non-adiabatic response in quantum and classical systems. *Physics Reports*, 697:1–87, 2017. doi: 10.1016/j.physrep.2017.07.001.
- [9] Jeeva Anandan and Yakir Aharonov. Geometry of quantum evolution. *Physical Review Letters*, 65:1697–1700, 1990.
- [10] Romeo Brunetti, Klaus Fredenhagen, and Rainer Verch. The generally covariant locality principle: A new paradigm for local quantum physics. *Communications in Mathematical Physics*, 237:31–68, 2003. doi: 10.1007/s00220-002-0722-0.
- [11] Marek J. Radzikowski. Micro-local approach to the hadamard condition in quantum field theory on curved space-time. *Communications in Mathematical Physics*, 179:529–553, 1996.
- [12] Romeo Brunetti and Klaus Fredenhagen. Microlocal analysis and interacting quantum field theories: Renormalization on physical backgrounds. *Communications in Mathematical Physics*, 208:623–661, 2000. doi: 10.1007/s002200050004.
- [13] Stefan Hollands and Robert M. Wald. Local wick polynomials and time ordered products of quantum fields in curved spacetime. *Communications in Mathematical Physics*, 223:289–326, 2001. doi: 10.1007/s002200100540.
- [14] B. S. Tsirelson. Quantum generalizations of Bell’s inequality. *Letters in Mathematical Physics*, 4:93–100, 1980. doi: 10.1007/BF00417500.
- [15] Leonid Mandelstam and Igor Tamm. The uncertainty relation between energy and time in non-relativistic quantum mechanics. *Journal of Physics USSR*, 9:249–254, 1945.
- [16] Xinsheng Tan, Dan-Wei Zhang, Zhen Yang, Ji Chu, Yan-Qing Zhu, Danyu Li, Xiaopei Yang, Shuqing Song, Zhikun Han, Zhiyuan Li, Yuqian Dong, Hai-Feng Yu, Hui Yan, Shi-Liang Zhu, and Yang Yu. Experimental measurement of the quantum metric tensor and related topological phase transition with a superconducting qubit. *Physical Review Letters*, 122:210401, 2019. doi: 10.1103/PhysRevLett.122.210401.
- [17] Min Yu, Pengcheng Yang, Musang Gong, Qingyun Cao, Qiuyu Lu, Haibin Liu, Martin B. Plenio, Fedor Jelezko, Tomoki Ozawa, Nathan Goldman, Shaoliang Zhang, and Jianming Cai. Experimental measurement of the quantum geometric tensor using coupled qubits in diamond. *National Science Review*, 7:254–260, 2020. doi: 10.1093/nsr/nwz193.
- [18] Ching-Yi Hsiang and collaborators. quantum\_memory\_data: Experimental data, analysis code, and figure reproduction. GitHub repository, 2026. URL [https://github.com/44hsiang/quantum\\_memory\\_data](https://github.com/44hsiang/quantum_memory_data). Accessed 24 August 2026.
- [19] Xuan Zhang, Xiao-Le Li, Jingjing Niu, Tongxing Yan, and Yuanzhen Chen. Dataset for engineered robustness for nonadiabatic geometric quantum gates. Zenodo dataset, 2025.
- [20] Christina Giarmatzi. NMN-tomo: Multi-time quantum process tomography on a superconducting qubit. GitHub repository, 2025. URL <https://github.com/Christina-Giar/NMN-tomo>. Accessed 24 August 2026.
- [21] Elija Perrier and collaborators. QDataSet: Quantum datasets for machine learning. GitHub repository, 2022. URL <https://github.com/eperrier/QDataSet>. Accessed 24 August 2026.
- [22] Qiskit Community. IBM Quantum Awards: Open Science Prize 2021. GitHub repository, 2022. URL <https://github.com/qiskit-community/open-science-prize-2021>. Accessed 24 August 2026.

- [23] Yincheng Liu, James R. Seddon, Tamara Kohler, Emilio Onorati, and Toby S. Cubitt. Robust lindbladian estimation for quantum dynamics, 2026. URL <https://arxiv.org/abs/2507.07912>.
- [24] Yincheng Liu, James R. Seddon, Tamara Kohler, Emilio Onorati, and Toby S. Cubitt. Hardware demonstration data for robust lindbladian estimation for quantum dynamics. Zenodo dataset, 2026.
- [25] Gal Ness, Manolo R. Lam, Wolfgang Alt, Dieter Meschede, Yoav Sagi, and Andrea Alberti. Observing crossover between quantum speed limits. *Science Advances*, 7:eabj9119, 2021. doi: 10.1126/sciadv.abj9119.